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## The Pink Rover Initiative: Launching the Next Generation of Space Innovators

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### Abstract

Globally, the space industry remains a male-dominated sector, lacking diversity and representation. Despite growing efforts to increase numerical representation in STEM, there are still considerable obstacles to gender equity, including implicit biases and structural barriers within both educational and professional environments. This lack of diversity in the space industry inhibits innovation, perpetuates inequality and fails to reflect the global nature of space exploration.

Monash Nova Rover (MNR) is a student-led engineering team that designs and builds analogue planetary rovers and has achieved 2nd place in international competitions. The team elected to paint their rover pink in 2023 to spark conversations about women in STEM, which has led to broader nuanced conversations about identity in STEM spaces and in the space industry. Through the Pink Rover Initiative, MNR aims to support diverse identities and self-expression in STEM. As the only Australian team to compete on the international stage, the team has continually proven that the MNR pink rover – and by extension the commitment to diversity, equity and inclusion it represents – belongs in STEM, and can achieve excellence.

Within MNR, the Pink Rover Initiative has built a culture of inclusion and increased sense of belonging, empowering members to thrive. The Pink Rover Initiative provides a safe and open platform to discuss deep rooted issues and is a communal responsibility that everyone on the team shares; this has ultimately brought the team closer together. Beyond MNR, the initiative sparks curiosity and engagement in STEM, guiding young students through hands-on mini-rover building workshops that combine technical skills with the promotion of inclusive values in STEM. In 2024, MNR engaged with over 24 schools, 2,000 students, and 16 outreach partners.

This paper presents the key insights of the Pink Rover Initiative, derived from a comprehensive evaluation of its impact within our team and the wider community. A mixed method approach was conducted to triangulate quantitative and qualitative data, ensuring that both measurable outcomes and human experiences are considered in this evaluation of social impact. Through surveys and interviews, the study aimed to assess Initiative reach, audience response, engagement, awareness of gender inequality, ability to recruit allies and ability to evoke reflection on internal biases.

With the space industry experiencing rapid expansion in 2025, there is a need to cultivate a workforce that is as diverse as the challenges it seeks to solve. A diverse and inclusive workplace not only advances gender equity but also provides access to a larger pool of scientific and creative thinking. The Pink Rover Initiative serves as an exemplary model for how meaningful change through provided platforms for nuanced conversation and reflection on unconscious bias can be successfully implemented in both tertiary studies and industry to drive innovation.

### Acronyms

MNR - Monash Nova Rover

STEM - Science, Technology, Engineering, and Mathematics

DEI - Diversity, Equity and Inclusion

PCA - Principal Component Analysis

SCT - Social Cognitive Theory

SD - Standard Deviation

### 1. Introduction

#### 1.1 Objectives

The primary aim of this research project is to critically analyse and evaluate the impact of the Pink Rover Initiative - identifying areas of success and areas that are limited. This paper contributes to understanding of how targeted initiatives can address unconscious bias in STEM through meaningful conversations, and can inform the design of future

programs. Following a literature review, section 2 presents the methodology for the evaluation of this initiative, section 3 presents results and discussion, and section 4 presents key conclusions.

## *1.2 Theory of Change*

By colouring a rover pink and positioning it at the heart of a renowned student robotics team, Monash Nova Rover (MNR), the rover transforms into a vehicle for change and sparks conversations about women and self-expression in STEM thus increasing inclusion, awareness and allyship.

## *1.2 Literature Review*

### *1.2.1 Gender Disparities in STEM*

In recent decades, the underrepresentation of women in STEM has been thoroughly researched and understood. According to the 2023 STEM Equity Monitor [1], 37% of students enrolled in university STEM degrees are women and 15% of qualified STEM jobs are occupied by women. Numerous factors contribute to these disparities, including societal stereotypes, implicit biases, and structural barriers within every stage of the educational and professional pipelines. Gender stereotypes are not only descriptive but also prescriptive [2], and thus undermine women's participation in fields that are stereotypically masculine. This barrier socially positions STEM to be perceived by some women as an intimidating and alien field, creating unequal opportunities to explore STEM.

According to the Australian Department of Education [3], boys are exposed to and encouraged to pursue careers in STEM, while girls are largely directed towards fields such as nursing and beauty. It was reported in the 2023 STEM Equity Monitor [1] that 'Girls' confidence in STEM subjects is generally lower than boys and falls as they get older.' This disparity is a reflection of the gender coding that permeates throughout society and negative stereotypes around STEM and gender, which serves to discourage young girls from exploring STEM fields. Given the correlation between early attitudes towards careers and subsequent career choices, it is essential that we provide equal opportunities for young girls to engage with and discover the potential of STEM.

Intersectionality plays a crucial role in shaping STEM participation as individuals may experience direct or indirect discrimination as a result of their

gender, race, ethnicity, socioeconomic status and sexual orientation [4]. Recognising and addressing the multifaceted nature of gender representation in STEM is essential for developing effective strategies to promote diversity, equity, and inclusion (DEI) within these fields.

### *1.2.2 Effects of Anti-Femininity Bias*

While decades of scholarship show the oppression of women by the enforcement of patriarchal gender norms, little research has explored the ways in which masculinity receives preferential treatment over femininity [5]. What must be challenged is why femininity is devalued in the heteropatriarchal masculine social context under which current western society was formed. Femininity presents as both a badge of identity and a barrier to entry in STEM. It can be argued that the masculine culture that characterises STEM and expressions of anti-femininity could contribute to the underrepresentation of women in STEM.

In order to fulfil their psychological needs, individuals identify with social groups. Social identity theory identifies two main motives for social identification: to gain positive self-esteem or self enhancement [6] and to reduce a sense of uncertainty about the self [7]. In line with social identity theory, women find it challenging to identify with STEM because the social environment provides a variety of signals that women do not belong to STEM and do not embody the prescribed STEM identity [8].

Betz and Sekaquaptewa [9] showed middle school girls magazine-style interviews featuring the same woman as either a gender-neutral or feminine STEM role model (e.g., dark clothes vs. pink clothes and makeup), both celebrated for STEM success. All participants linked the feminine role model to lower maths interest, ability, and short-term success expectations. The results of the study conclude that STEM-disidentified girls saw STEM as unattainable, suggesting that their demotivation was related to the perceived unlikely combination of femininity and STEM success. This reflects internalised anti-femininity bias and its effects on self technical efficacy. However, to avoid reinforcing stereotypes, STEM should be introduced in gender-neutral ways. A wider picture can be drawn from this study; that STEM is labelled unfeminine and costs the technical self-efficacy of some individuals who do not identify with STEM.

There is evidence in research showing that women in STEM minimise their feminine appearance, for example not wearing make-up when they usually would elsewhere, whilst also eschewing feminine traits, behaviours, or goals perceived as emotional or domestic [10]. Cultures that devalue femininity can also lead women to distance themselves from and criticise other women, especially feminine women [11, 12]. Such practices reinforce that femininity and STEM are incompatible, creating harmful environments for the women involved that can lead to isolation and potential exit from STEM fields. [13, 14].

Social Cognitive Theory (SCT) suggests individuals are not entirely shaped by their environment nor fully autonomous, but actively influence their experiences based on self-efficacy [15]; people tend to avoid activities they feel exceed their abilities. The interaction between self-efficacy and environment highlighted by SCT helps explain women's participation and retention in STEM. For example, changing stereotypical objects in a computer science classroom (e.g., replacing Star Trek posters with nature posters) significantly increased women's interest in the field [16]. This research examines SCT through the Pink Rover Initiative, which seeks to boost technical self-efficacy and challenge stereotypes within the MNR team and the broader STEM community.

### *1.2.3 Research Gaps and STEM initiatives today*

Governments globally have acknowledged the need for leadership and policies to ensure the establishment and achievement of gender equity since the United Nations World Conference on Women in 1975 [17]. While many organisations and institutions have developed strategic plans designed to encourage girls and women to pursue education and careers in STEM, of 337 women-in-STEM initiatives and programs that were mapped as part of the Australian Decadal Plan for Women in STEM, only seven had been formally evaluated [18].

The WISET advisory group was formed in 1995 to offer strategies that promote greater participation in STEM careers, approaching their task by asking “what is it about the environments of science, engineering and technology, and society’s perception of them, that they do not attract and keep girls and women?”. Over twenty years later, we have come part of the way in answering this question, but core

challenges remain [18]. The 2018–2019 Australian Federal Budget signalled strong political support for improving girls’ and women’s participation in STEM through funding for the Decadal Plan, a 10-year roadmap to attract, retain, and support women and girls by addressing cultural and systemic barriers [19]. While WISET and the Decadal Plan for Women in STEM were published almost 20 years apart, both make similar recommendations, especially related to the importance of data collection and identification of best practice in programmes and initiatives designed to support girls and women in STEM within both public and private sectors [18]. In 2022, Science Minister Ed Husic ordered a review of federally funded programs designed to encourage women into science and technologies after \$100 million over the past decade failed to shift the dial [20]. While interventions targeting numerical representation are important, they do not necessarily address the underlying cycle of recruiting and retaining more women in STEM [16]. An independent panel conducted a year-long investigation into the most effective ways to increase diversity in STEM; the Australian Government’s Pathway to Diversity in STEM Review [21] makes final recommendations on the need to address unconscious bias, measure the success of initiatives, and ‘work to shift pervasive perceptions and stereotypes about who can, and should, work in STEM jobs’ [21].

The paper “But You Don’t Look Like A Scientist!” examined whether subtle variations in feminine appearance erroneously convey a woman’s likelihood of being a scientist [22]. The paper evaluated the #iLookLikeanEngineer campaign, a marketing strategy by companies and universities to visibly highlight diversity in their STEM fields. The results provide evidence that counter-stereotype campaigns can alleviate pressure on women in STEM to suppress their femininity.

From 2018 to 2024, Professor Lisa Harvey-Smith acted as Australia’s first Women in STEM Ambassador, leading programs to contribute research, tools and resources to support women and girls to succeed in STEM [23]. This program saw the launch of Future You, a national initiative that addresses stereotypes about the STEM field and aims to increase engagement in 8-12 year olds [24]. Moreover, STEMpals, an initiative launched at Monash University’s The Generator, connects students aged 10-13 with STEM mentors through

themed letter exchanges to challenge stereotypes and promote diversity in science [25]. In 2024, 745 students across Australia participated, with drawing activities revealing a shift from students drawing traditional scientist imagery to imagining diverse teams and even themselves in STEM roles [25]. While these initiatives provide exemplary examples of approaches to addressing gender gaps in STEM, the evaluation of such programs is imperative for identifying successes and limitations.

## 2. Methodology

### 2.1 Researcher Positionality

The research team comprises six undergraduate engineering students who are all members of the MNR Pink Rover Committee, including the current Pink Rover Chair, as well as one of the founders of the initiative, who is an alumnus of the team. All researchers thus have a personal interest in the evaluation of the initiative, bringing strong contextual knowledge of the initiative's mechanisms and a shared commitment to the work. All efforts have been made to manage potential bias and ensure an impartial analysis of evaluation results, with reviews of the paper by external supervisors being undertaken to address the influence of this positionality.

### 2.2 The Pink Rover Model

MNR is a student-led engineering team that designs and builds analogue planetary rovers and has achieved national and international competition titles. The team encompasses approximately 100 students from academic backgrounds in engineering, science, computer science, business and design. The Pink Rover Initiative was founded by the team in 2022 with the decision to paint their flagship rover hot pink to 'start conversations about women in STEM', and the Pink Rover Campaign was subsequently launched at the beginning of 2023. This initiative symbolises the team's effort to address the nuanced social stereotypes associated with it, and centres on three pillars:

- Empower through Education
- Bring Visibility and Foster Engagement
- Address Barriers and Bias

Traditionally, the colour pink is associated with femininity; conversely, stem fields are associated with masculine traits. The unlikely intersection of these two notions - the colour pink and STEM - juxtaposes with the absence of pink in STEM, and challenges unconscious perceptions of who can enter and thrive

within this space. The theory of change, informed by SCT, anticipates a broad expected outcome of introducing change to an environment but does not imply a rigid hypothesis [26].

Since Pink Rover's launch, foundational messaging has evolved to match the Initiative's growth. The title shifted from *Campaign* to *Initiative*, reflecting a move from short-term awareness to a sustained, strategic program. The slogan also broadened from "sparking conversations about women in STEM" to "sparking conversations about self-expression, opportunity, and diversity in STEM." The updates aim to include both non-women that experience challenges related to feminine self-expression, and non-feminine women.

With each design cycle, MNR not only iterates on rover and payload designs but also experiences team turnover and recruitment. At the start of each cycle, the team collectively reviews internal and external feedback, refining messaging through a community-driven approach that values diverse perspectives. This alignment ensures the Initiative's messaging reflects both community needs and the evolving team identity.

### 2.3 Systematic Examination of Research Question

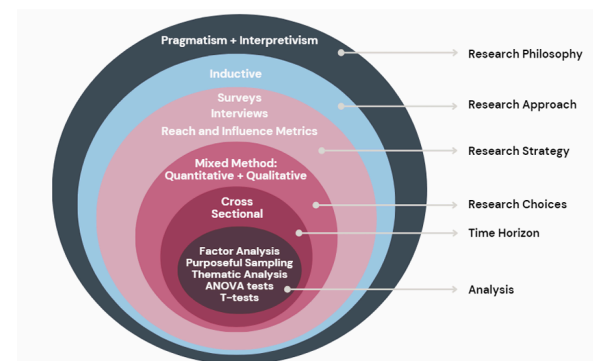


Fig. 1. The Research Onion

The 'Research Onion' was used to provide a systematic examination of the research question. This model provides a structured framework for conducting research, guiding researchers through different layers of methodology, from philosophy to techniques and procedures [27]. The research adopts a combination of pragmatism and interpretivism. Pragmatism focuses on practical outcomes, emphasising real-world solutions to the underrepresentation of women in STEM through the Pink Rover Initiative. This approach values both

qualitative and quantitative methods, allowing for flexibility in addressing complex challenges. Interpretivism complements this by seeking to understand the subjective experiences and meanings that individuals attach to the Initiative. The Initiative seeks to raise awareness of unconscious bias in STEM, and the research aims to explore shifts in attitudes and perceptions. This requires open-ended, data-driven exploration, rather than testing predefined hypotheses. The Theory of Change provides a broad expected outcome but is not confined to a purely deductive approach where a rigid hypothesis is tested [26]. Instead, the Theory of Change serves as a guide for understanding how change might occur while remaining open to new findings. Combining qualitative and quantitative data, such as experiences and cultural differences, the inductive approach enables the development of broader themes. There is no established theory on how an initiative like Pink Rover affects perceptions of STEM. Instead of testing an existing framework, the research will uncover new patterns and insights.

Mixed method research integrates both quantitative and qualitative data collection and offers significant benefits for social impact research. Quantitative methods provide the ability to measure and analyse numerical data, yielding statistical evidence of trends, patterns, and correlations. This can help establish the scope and scale of the social impact. Conversely, qualitative methods offer in-depth insights into personal experiences, motivations, and contextual factors, enriching the understanding of complex social phenomena. Triangulating these two approaches enhances the validity and reliability of findings by cross-verifying data from different sources and ensures that both measurable outcomes and human experiences are considered in the evaluation of social impact.

Participation metrics have been extracted to quantify engagement with Initiative activities, including attendance at workshops, events, and online discussions as well as capturing the Pink Rover Initiative's digital footprint. These metrics provide tangible evidence of the Initiative's reach and effectiveness in fostering dialogue and engagement within MNR and the broader community. Social media analytics were utilised to monitor the reach and engagement of the Initiative's online content. This data offers insights into the Initiative's digital footprint and its ability to disseminate information

and spark conversations. Prior to the commencement of data collection, the evaluation design went under an ethics approval process through the Monash University Ethics Review Manager for all interview questions and surveys - MUHREC ID 42935.

A cross-sectional time horizon involves collecting data at two single points in time (mid 2024 and mid 2025) to provide a snapshot of the current situation at each of those points. Data collected in mid 2024 was collected 1.5 years after the launch of the Pink Rover Initiative from participants associated with Monash University Engineering student teams, as well as external stakeholders of the Pink Rover Initiative to provide a snapshot of the Initiative's immediate impact. Data collected in mid 2025 was obtained from the same participant groups, as well as expanding to include students from other national and international engineering student teams. Further data collection enables a more comprehensive analysis of how the impact of the initiative has evolved over time, as well as allowing new metrics, such as reach to other engineering teams, to be created and analysed.

Table 1. Data types and collection methods

| Data Type |              | Collection Method |           |
|-----------|--------------|-------------------|-----------|
|           |              | Survey            | Interview |
|           |              | 2024, 2025        | -         |
|           | Quantitative | 2024, 2025        | -         |
|           | Qualitative  | 2025              | 2024      |

For quantitative data analysis, T-tests and ANOVA tests were used to compare differences in attitudes and perceptions between groups, while Principal Component Analysis (PCA) helped reduce dimensionality and identify key factors influencing responses and confirming reliability of clusters of questions to measure constructs. For qualitative data processing, thematic analysis was conducted to analyse interviews and short-answer survey questions to uncover recurring themes. Purposeful sampling ensured that participants were selected based on relevance to the Pink Rover Initiative and student team experience, allowing for the generation of insights into the Initiative's impact.

For data analysis in 2025, the 2024 survey was re-distributed with tailored questions based on

outcomes from 2024 PCA testing to augment the existing research.

## 2.4 Interviews

Interviews conducted for this research use a semi-structured format to explore participants' experiences with the Pink Rover Initiative, focusing on its successes, limitations, and impact. The discussions aim to explore interviewees' understanding of the objectives, messaging, and what underlying mechanisms unlock reflection of unconscious bias and raise awareness. This approach aims to facilitate open and candid discussions while ensuring key topics related to the research objectives are systematically addressed. Refer to Appendix B for interview questions.

The areas that can be further explored from the survey in the interviews are:

- Student Team Experience
- Bias, Gender identity and self-expression
- Allyship, Responsibility and Awareness of gender inequality
- Response to the Pink Rover and Gender symbolism
- Successes and limitations of the Pink Rover Initiative

Participants include MNR team members and external stakeholders involved in Pink Rover events or Initiative development. Interview questions - demographic, open-ended, and behavioral - explore their experiences with the initiative and in STEM. The interviews aim to capture personal perspectives on the Initiative's goals, messaging, and impact on perceptions of STEM and unconscious bias. Topics include initial reactions to Pink Rover, involvement in early development, changes in team dynamics or community dialogue, and shifts in perceptions of STEM and bias. Confidentiality and ethical conduct are prioritised, with all participants provided with explanatory statements, consent forms, and the option to withdraw at any time.

## 2.5 Surveys

The 2024 and 2025 surveys aim to identify the factors that contribute to the limitations or success of the Pink Rover Initiative. Both surveys include statements that participants can agree or disagree with on the Likert seven point scale, refer to Appendix A for 2024 survey questions and Appendix C for 2025 survey questions. There are a few open-ended questions included to allow participants to expand on

their answers or provide additional insights for qualitative data collection. While the 2024 interviews were designed to capture in-depth personal perspectives, the qualitative survey questions were designed to provide participants with an opportunity to provide short, honest responses and comments to capture common patterns among respondents. The survey questions were designed based on three constructs to be investigated: individual perceptions of STEM, engineering student team experience and Pink Rover Initiative experience. The survey results were used to measure the effects of exposure to the initiative and gender on these constructs. The surveys target MNR team members, other domestic and international engineering student teams and external stakeholders of the Pink Rover Initiative.

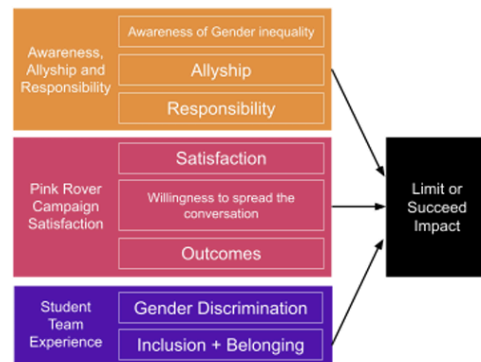


Fig. 2. Survey Constructs Diagram

Key focus areas of the survey include participants' feelings of inclusion and team belonging, gender discrimination, familiarity with the Pink Rover Initiative, their impressions of the Initiative's messaging and goals, and any changes in their attitudes toward gender diversity in STEM as a result of the Initiative. The survey also measures participants' awareness of bias within the engineering community and broader STEM fields and investigates if Pink Rover prompted self reflection. The survey was designed with clear, relevant language to ensure ease of comprehension. Pre-survey testing with target participants helped refine the questions for clarity and consistency. To maximise participation, the survey was distributed via email and online platforms, with snowball sampling encouraging participants to share it further within their networks. The survey ensures confidentiality and ethical standards with participant anonymity, allowing participants to skip qualitative questions, and providing clear instructions and consent details at the commencement of the survey.

## 2.6 Analysis Approach

SPSS Statistics was used to analyse quantitative data from surveys. The quantitative data analysis will explore the main areas:

1. Examining potential correlations between participants' demographic factors, such as, gender, student team experience, awareness of the Pink Rover Initiative and their responses to the Initiative.
2. Investigating responses to the Pink Rover Initiative over time using data collected in 2024 and 2025, exploring Pink Rover's ability to provoke reflection of unconscious bias and raise awareness of gender inequality in STEM.
3. Differences between participants exposed to the Pink Rover Initiative and those not exposed, student team experience, awareness of gender disparities in STEM and allyship.
4. Differences between men and non-men in student team experience, awareness of gender disparities in STEM and allyship.
5. Differences between MNR members and Non-MNR student team members in student team experience, awareness of gender disparities in STEM and allyship.

A Principal Component Analysis (PCA) was conducted on the sixteen 2024 student experience survey questions to reduce dimensionality and identify underlying patterns in the data. PCA calculates the correlating variation among a set of observed variables to identify underlying dimensions/constructs by obtaining the covariance matrix of the variables, and then its eigenvectors and the corresponding eigenvalues [28]. Cronbach's alpha was used to examine the internal validity of the items for each component. It is a statistic used to measure how well a set of questions in a survey that are supposed to measure the same construct or concept, do so consistently. All constructs scored higher than 0.7 indicating high reliability. To assess whether respondents' gender or awareness of/involvement with Pink Rover had a significant effect on each of the constructs identified from the PCA, independent samples t-tests were performed for each survey question and for the sum of all items within each dimension. Effect sizes were assessed using Cohen's *d*, where 0.2, 0.5, and 0.8 indicated a small, medium and large effect, respectively. To further evaluate the effects and interactions of the Pink Rover Initiative

and respondents' gender, two-way ANOVA tests were performed.

In addition to quantitative analysis, qualitative data from the interviews and 2025 survey short answer questions were analysed using thematic analysis. Thematic analysis involves coding the data to identify recurring themes, patterns, and key insights related to the research questions and objectives. Themes can subsequently be organised into meaningful categories to facilitate interpretation and synthesis of qualitative findings.

## 3. Results and Discussion

This section outlines reach and influence metrics of the Pink Rover Initiative and discusses the demographics of the participants to provide context for the findings. An analysis of the quantitative survey results and qualitative insights from the interviews and short-answer survey questions will subsequently be presented, underscoring key findings. Finally, the discussion explores the relevance and impact of these observations within the framework of this research and proposes insights applicable to future inclusion initiatives.

### 3.1 Pink Rover Initiative Reach and Influence Metrics

This section summarises the activities of the Pink Rover Initiative since its launch in February 2023 and provides the reach of the Initiative through participation metrics, reach in media and influence within industry, government and community.



Fig. 3. Pink Rover June 2023 at the University Rover Challenge, Utah

#### 3.1.1 Empower through Education

This pillar of the initiative focuses on outreach and the introduction of robotics to as many young students as possible. Since the launch of the Initiative, MNR has participated in 88 outreach workshops or events. In MNR-led outreach workshops, team members teach primary and secondary school students how to build small Mars rovers. Events

include giving presentations to students, or hosting stalls at shows such as the Avalon Airshow, Australia Manufacturing conference, Space Forum or Monash University Open Day. Through outreach efforts since the launch of Pink Rover, approximately 5500 students have been reached through MNR workshops and expos. Many more community members have been reached through MNR attended or hosted events and competitions, as well as through the team's social media presence.

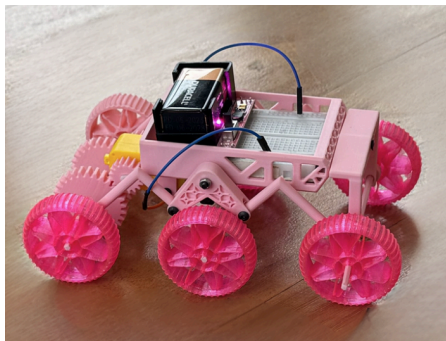


Fig. 4. MNR Outreach Mini Rover Kit

### 3.1.2 Bring Visibility and Foster Engagement

This pillar aims to highlight the talented women already excelling in STEM, and to create events and spaces that foster community growth and support networks. Additionally, it seeks to provide women role models through media and storytelling, affirming that exposure is essential for inspiring the next generation of space innovators.

The 2023 Launch of Pink Rover attracted an audience of 300 people. To bring visibility and foster engagement, a women in STEM panel night was held by MNR in 2023, which consisted of 11 panelists and 4 expert speakers, attracting an audience of 153 people.. The first panel 'Future Leaders in STEM' included five young women who held technical leadership roles in prominent engineering student teams at Monash. The second panel, 'Women in Space and Robotics' included 4 accomplished women from the industry. The aim of these discussions was to understand the experiences of women in STEM and learn from their technical expertise. This event aimed to highlight the talented women already in STEM and offer diverse and empowering role models to other women studying STEM or looking to pursue it. In 2024, MNR hosted a second panel night; 'Human-Machine Interaction and Augmented

Reality' with eight panellists comprising industry leaders, academics and MNR members.



Fig. 5. 2023 Women in STEM Panel Night

The Initiative also brings visibility to the talented women on the team through meaningful messaging and storytelling on the team's social media platforms, as well as the implementation of a promotional strategy to further spread the conversation about Pink Rover. As of September 2025, MNR accounts have 7500 followers across Instagram, Facebook and LinkedIn, reaching many more accounts through social media platforms. Pink Rover conversations have also been spread following coverage by news outlets and inclusion in 19 articles, as well as discussed in coverage of MNR-attended competitions such as ARCh 2025. Most notably, Pink Rover was featured in ABC Behind the News, a Mecca M-power article and in a Nature article.

Furthermore, MNR has been invited to speak at five conferences since the launch of Pink Rover, most notably the Global Institute for Women's Leadership: Diversity at the Frontier Conference. The team has also had many opportunities to engage with policy and government whilst receiving overwhelming support from the Australian Space Agency. Through this Initiative, the team has had the opportunity to contribute to the Australian Robotics strategy. Pink Rover efforts have garnered attention from many in the space industry and government, including the Vice Chancellor of Monash University, the Deputy Director of the Mars Exploration Program, Robotics Group Australia, the Dean of Engineering, the Chief Scientist of Victoria, and the Minister of Defence, among others.

### 3.1.3 Address Barriers and Bias

To address barriers and bias in STEM, the team organised workshops on campus for university students to learn more about structural inequality and systematic bias that contributes to the underrepresentation of women in STEM. These were

led by external professionals from STEM Sisters. Bias is a construct that the Initiative aims to target through the image of the Pink Rover itself. The Pink Rover is a polarising image and attracts attention due to its antithetical nature; an inherent Pink Rover mechanism. As the question ‘why pink?’ is so often prompted by the Pink Rover image, it provokes self-reflection and challenges previous ways of thinking about STEM, including perceptions of who belongs in it. This evaluation aims to investigate if the Pink Rover is, in fact, successful in sparking those conversations and raising awareness around unconscious bias in STEM.

### 3.2 Demographics

Table 2. 2024 Survey Demographics

|                                                                 | Group             | N  |
|-----------------------------------------------------------------|-------------------|----|
| <b>Are you in an Engineering Student Team?</b>                  | Yes               | 91 |
|                                                                 | No                | 3  |
| <b>Were you aware of/involved with the Pink Rover Campaign?</b> | Yes               | 63 |
|                                                                 | No                | 31 |
| <b>Gender</b>                                                   | Women             | 31 |
|                                                                 | Men               | 41 |
|                                                                 | Non-binary        | 3  |
|                                                                 | Gender fluid      | 1  |
|                                                                 | Prefer not to say | 18 |
| <b>What student team are you in?</b>                            | MNR               | 24 |
|                                                                 | Other             | 67 |

Table 3. 2025 Survey Demographics

|                                                                   | Group | N   |
|-------------------------------------------------------------------|-------|-----|
| <b>Are you in an Engineering Student Team?</b>                    | Yes   | 212 |
|                                                                   | No    | 5   |
| <b>Were you aware of/involved with the Pink Rover Initiative?</b> | Yes   | 157 |
|                                                                   | No    | 55  |
| <b>Gender</b>                                                     | Women | 71  |
|                                                                   | Men   | 105 |

|                                      |                   |     |
|--------------------------------------|-------------------|-----|
|                                      | Non-binary        | 3   |
|                                      | Gender fluid      | 2   |
|                                      | Prefer not to say | 36  |
| <b>What student team are you in?</b> | MNR               | 59  |
|                                      | Other             | 153 |

Table 4. 2024 Interview Demographics

| Gender                   | Group                | N |
|--------------------------|----------------------|---|
| <b>Women, Non-binary</b> | MNR                  | 9 |
|                          | External Stakeholder | 6 |
| <b>Men</b>               | MNR                  | 6 |
|                          | External Stakeholder | 2 |

### 3.3 Findings

#### 3.3.1 Principal Component Analysis

A Principal Component Analysis (PCA) was conducted on all survey questions to identify key factors.

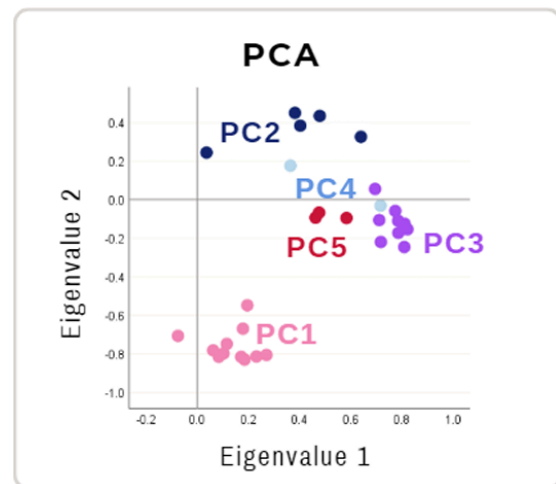


Fig. 6. PCA on the 2024 Survey Questions

Suitability of the 2024 data for PCA was indicated by a Kaiser-Meyer-Olkin (KMO) measure of 0.681 (moderate sampling adequacy) and a significant Bartlett's Test of Sphericity ( $\chi^2 = 1634.580$ ,  $p < 0.001$ ), confirming sufficient common variance among variables. As shown in Figure 6, five distinct clusters of questions emerged. To explore underlying

patterns further, a PCA was also performed on the survey's three sections. This approach enables clearer identification of specific constructs, improves statistical power, and enhances insights into variable relationships.

The first section addresses the student team experience, with PCA performed on 16 questions. The KMO measure was 0.876, indicating excellent suitability for factor analysis, while Bartlett's Test yielded a significant chi-square value ( $\chi^2 = 1192.15$ ,  $p < .001$ ), supporting the factorability of the correlation matrix. Two distinct dimensions emerged: PC1, "Feelings of Inclusion and Team Belonging," explained 48.22% of the variance (items EXP1 to EXP11), and PC2, "Discrimination and Self-Doubt," explained 18.37% (items EXP11 to EXP16). Together, these factors accounted for 66.59% of the total variance.

The second section, focusing on Pink Rover Initiative satisfaction, showed high PCA appropriateness with a KMO of 0.912 and significant Bartlett's test results ( $\chi^2 = 589.838$ ,  $p < 0.001$ ). A single component, PC3 "Pink Rover Initiative Satisfaction," was extracted, with strong loadings (0.722 to 0.894) explaining 68.19% of the variance.

The final section assesses awareness, allyship, and responsibility regarding gender equity in STEM. The data demonstrated PCA suitability with a KMO of 0.719 and a significant Bartlett's Test ( $\chi^2 = 185.110$ ,  $p < 0.001$ ). Two components were identified: PC4, "Awareness of Gender Inequalities in STEM," which accounted for 23.20% of the total variance, and PC5, "Allyship and Responsibility," which explained an additional 55.70%, resulting in a cumulative variance of 78.90%. This confirms the effectiveness of these components in capturing the data's underlying structure.

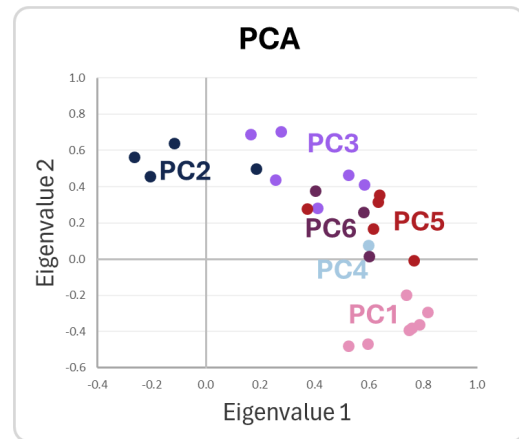


Fig. 7. PCA on the 2025 Survey Questions

Suitability of the 2025 data for PCA was indicated by a KMO measure of 0.727 (good sampling adequacy) and a significant Bartlett's Test of Sphericity ( $\chi^2 = 751.58$ ,  $p < 0.001$ ), confirming sufficient common variance among variables. As shown in Figure 7, six distinct clusters of questions emerged. To explore underlying patterns further, a PCA was also performed on the survey's four sections, as done to the 2024 sections.

The first section addresses the student team experience. The KMO measure was 0.919, indicating excellent suitability for factor analysis, while Bartlett's Test yielded a significant chi-square value ( $\chi^2 = 1001.79$ ,  $p < .001$ ), supporting the factorability of the correlation matrix. A single component, PC1 "Feelings of Inclusion and Team Belonging," was extracted, explaining 68.22% of the variance.

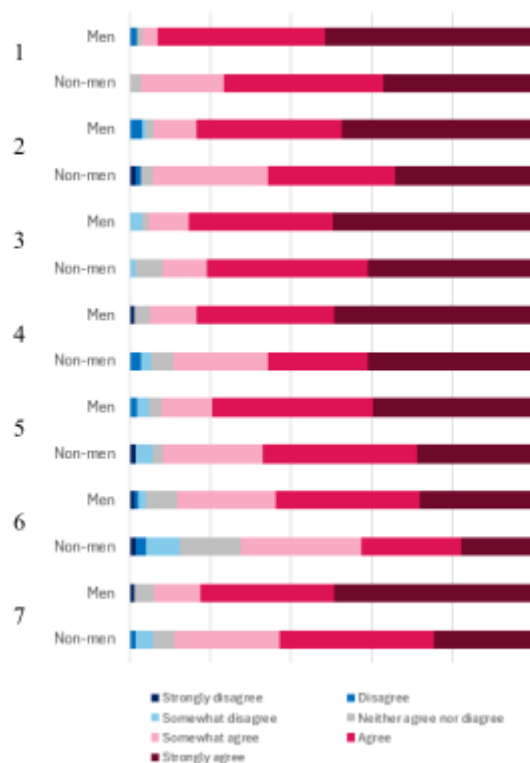
The second section, focusing on Experiences in STEM, showed good PCA appropriateness with a KMO of 0.776 and significant Bartlett's test results ( $\chi^2 = 657.30$ ,  $p < 0.001$ ). Three components were identified: PC2, "Bias, Gender Identity and Authenticity in the STEM Environment," which accounted for 19.06% of the total variance, PC3, "Allyship, Responsibility and Awareness," which explained 31.993% of the total variance, and PC4, "Ability to Create Change," accounting for 10.01% of the variance, resulting in a cumulative variance of 61.06%. Due to the affinity of PC3 and PC4, these components will be analysed together under the same grouping: Allyship, Responsibility and Awareness.

The third section focuses on Gender Symbolism within the Pink Rover Initiative. The KMO measure was 0.573, indicating acceptable suitability for factor analysis, however Bartlett's test proved the results to

be of low significance ( $\chi^2 = 5.889$ ,  $p < 0.117$ ). One component, PC5 “Response to the Pink Rover Initiative and Gender symbolism,” was still extracted, accounting for 47.34% of variance. Due to its low significance, the questions in this component will be analysed separately. Any similar results between all questions can then be used to analyse the component as a whole.

The final section assesses satisfaction with the Pink Rover Initiative. The data demonstrated PCA suitability with a KMO of 0.791 and a significant Bartlett's Test ( $\chi^2 = 259.052$ ,  $p < 0.001$ ). A single component was identified, PC6 “Pink Rover Initiative Satisfaction,” accounting for 60.82% of variance.

### 3.3.2 Feelings of Inclusion and Team Belonging



Regarding my student team experience, I feel...

1. Trusted by my teammates.
2. Valued on my team.
3. Supported by my teammates.
4. Comfortable asking my teammates for help.
5. Heard and listened to on my team.
6. Strong confidence in my technical ability.
7. Authentic to myself on my team.

Fig. 8. 2025 Gender differences in Feeling of Inclusion and Belonging in Teams

The above statements aim to measure feelings of inclusion and team belonging by analysing an individual's perception of being respected, supported and able to contribute authentically and meaningfully within their team. Research indicates that when members feel genuinely included, valued and psychologically safe to contribute, their willingness to engage with and invest effort into their teams improves significantly, enhancing team performance [29]. This highlights the importance of fostering this team culture for all members, where diversity is not only represented but actively valued.

An independent sample t-test on gender revealed overall high scores across all seven questions, with men generally reporting higher mean scores. For instance, men expressed slightly higher feelings of being valued ( $M = 6.18$ ,  $SD = 1.091$ ) compared to non-men ( $M = 5.88$ ,  $SD = 1.142$ ), though this difference was not statistically significant ( $p = 0.057$ , Cohen's  $d = 0.21$ ). Statistically significant ( $p < 0.037$ ) differences were observed for men versus non-men for trust (men:  $M = 6.38$ ,  $SD = 0.88$ ; non-men:  $M = 6.11$ ,  $SD = 0.83$ ), comfort asking for help (men:  $M = 6.25$ ,  $SD = 0.98$ ; non-men:  $M = 5.88$ ,  $SD = 1.25$ ), technical confidence (men:  $M = 5.74$ ,  $SD = 1.16$ ; non-men:  $M = 5.15$ ,  $SD = 1.38$ ), and authenticity (men:  $M = 6.23$ ,  $SD = 1.00$ ; non-men:  $M = 5.70$ ,  $SD = 1.11$ ) all with large to very large effect sizes (Cohen's  $d = 0.86, 1.10, 1.26$  and  $1.05$  respectively). For other statements, no statistically significant differences were found. Generally, non-men also showed greater variability, with some participants reporting lower feelings of value, comfort in asking for help and confidence in technical ability.

Of particular interest, non-men reporting far lower technical confidence coupled with a hesitance to seek help indicates they experience or perceive higher levels of negative judgement for not being at the expected level of competency. Alternatively, non-men may hold themselves to higher standards than their male peers. Overall, this response pattern and combined large effect size indicates that men feel more positively about their student team experience, feeling more empowered to contribute to, learn from and present authentically within their teams compared to non-men.

An analysis of these same sentiments between MNR and non-MNR participants showed a higher perception of inclusion and belonging in MNR team

members across all seven categories with an average of  $M = 6.28$ ,  $SD = 0.118$  compared to  $M = 5.92$ ,  $SD = 0.096$  with large Cohen's  $d$  values ( $>1$ ). Statistical significance ( $p < 0.05$ ) was found in feelings of trust, value, support, comfort asking for help and authenticity. Most notably, MNR members felt much more empowered to ask for help with less fear of judgment (MNR:  $M = 6.54$ ,  $SD = 0.68$ ; Non-MNR:  $M = 5.93$ ,  $SD = 1.24$ ;  $p = 0.00$ , Cohen's  $d = 1.11$ ).

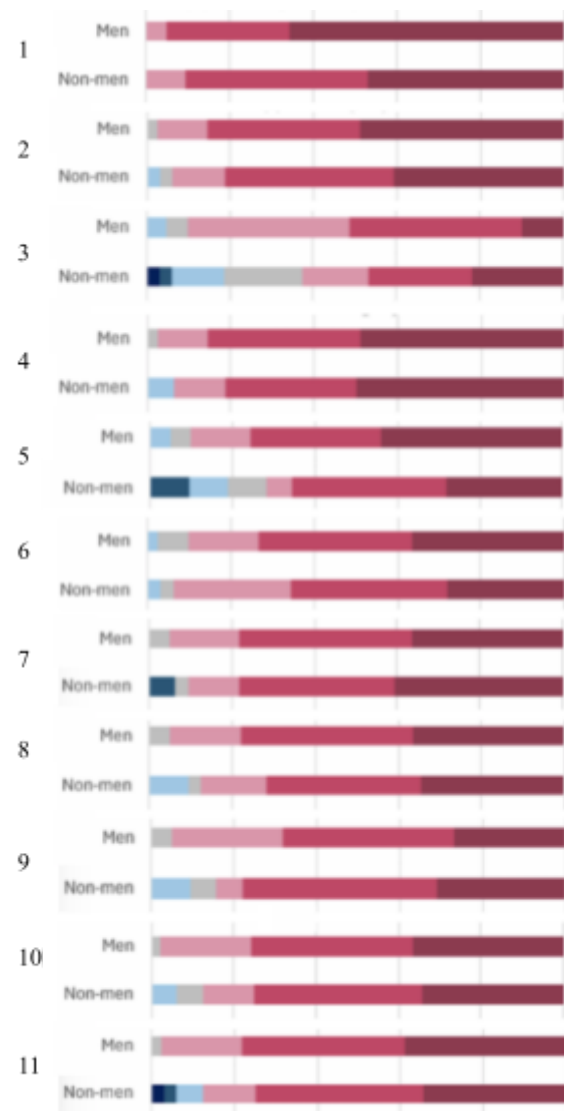
These findings suggest that MNR perpetuates a culture of inclusivity and safety leading to higher feelings of team affinity. Given that the Pink Rover Initiative was founded by and is proudly championed by MNR, it is likely that the initiative has heightened awareness of gender-based discrimination and the importance for diversity in STEM within the team's culture. By placing inclusivity and diversity at the forefront of MNR members' minds and actions, the initiative may in part be responsible for leading to more inclusive behaviours and thus a stronger sense of belonging among members.

A t-test performed between participants aware of or involved in the Pink Rover Initiative versus those who were not revealed no substantial differences on most measures of team inclusion and belonging. The only statistically significant difference emerged for the statement "I feel comfortable asking my teammates for help" where Pink Rover-aware students reported greater comfort ( $M = 6.21$ ,  $SD = 1.00$ ) compared to those not aware ( $M = 5.78$ ,  $SD = 1.45$ ;  $p = .045$ ,  $d = 0.38$ ). Although not statistically significant, Pink Rover-aware students also tended to report feeling slightly more authentic on their team ( $M = 6.10$  vs.  $5.76$ ,  $p = .105$ ,  $d = 0.30$ ). While numerical data did not show strong correlations between Pink Rover awareness and positive feelings of inclusion, participant quotes indicate there is a positive influence.

*"The pink rover campaign genuinely made me feel like I'd be valued and respected in the team." - Female Engineer on MNR (2025)*

*"The rover being pink indicated to me that the team was comfortable with femininity and those conversations which made me feel welcome, a toxic 'boys club' would never have a pink rover." - Female MNR Member (2025)*

While little significant differences were found with independent t-tests, a two-way ANOVA test between Pink Rover awareness and gender revealed insights on the interaction of these two categories. Importantly, a pattern emerged for authenticity ( $p = .027$ ,  $\eta^2 = .03$ ), suggesting that Pink Rover awareness was particularly beneficial for non-men, who reported substantially higher authenticity when aware ( $M = 5.93$ ,  $SD = 0.95$ ) compared to when not aware ( $M = 5.05$ ,  $SD = 1.31$ ). No other results were found to be significant, however, non-men reported a higher mean score for comfort asking for help when aware ( $M = 6.09$ ,  $SD = 1.03$ ) versus not ( $M = 5.26$ ,  $SD = 1.59$ ).



1. I enjoy working with my student team.
2. I feel supported by my teammates.
3. I feel strong confidence in my technical ability.
4. I feel comfortable asking my teammates for help.
5. I feel included and part of the team all the time.

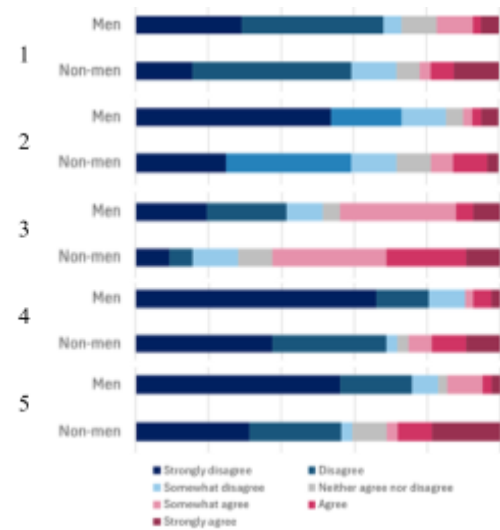
6. I feel heard and listened to on my team.
7. I feel I am able to contribute to the team.
8. I feel valued on my team.
9. I feel comfortable to make mistakes on my team.
10. I feel trusted by my teammates.
11. I feel authentic to myself in STEM spaces.

Fig. 9. 2024 Gender differences in Feeling of Inclusion and Belonging in Teams

The 2024 research Pink Rover Initiative found similarly high levels of reported inclusion and belonging on average for non-men ( $M = 6.425$ ,  $SD = 1.22$ ) and men ( $M = 6.7$ ,  $SD = 0.86$ ) across a similar set of questions. Although not statistically significant ( $p = 0.188$ ), non-men reported lower technical confidence ( $M = 5.03$ ,  $SD = 1.661$ ) than men ( $M = 5.46$ ,  $SD = 0.925$ ). Similarly, both an independent sample t-test conducted on Pink Rover exposure as well as an ANOVA investigating gender and initiative exposure interactions saw a lower mean score for non-men with no statistical significance. While quantitative data found no measurable impact of the initiative, 2024 interview data from female engineering students revealed the initiative's ability to create a safer STEM environment for self expression and greater inclusion on an individual level, indicating the campaign's power to positively influence feelings of belonging for minority groups in STEM spaces.

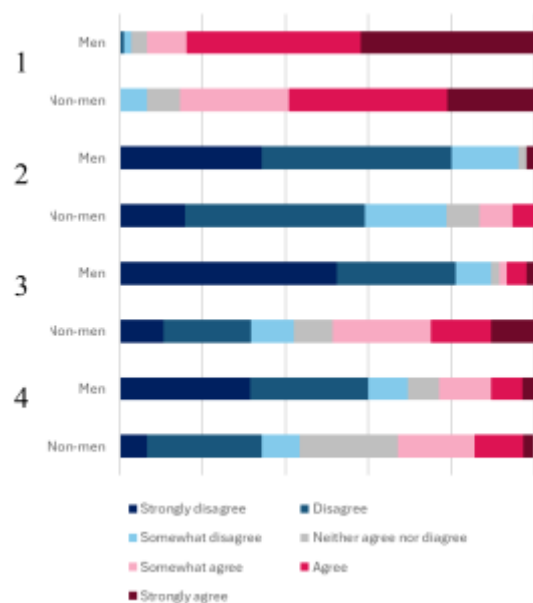
Compared to findings from the 2024 study on the Pink Rover Initiative, 2025 survey data shows a similar overall trend where all participants reported high levels of inclusion and belonging, while non-men consistently expressed lower levels of comfort, confidence and authenticity within their teams. In contrast to the 2024 study, where mostly non-statistically significant differences were found with gender and no measurable effects found for Pink Rover awareness, this study found several statistically significant gender differences with large effects sizes for a number of statements. Likewise, these results revealed a notable benefit for Pink Rover aware non-men in feeling authentic within STEM spaces. Together, these findings suggest that while the overall positive culture of team belonging has remained stable, the influence of gender and awareness of inclusivity initiatives such as Pink Rover may be becoming more pronounced over time, particularly for non-men.

### 3.3.3 Bias, Gender Identity and Authenticity in the STEM Environment



1. I have had negative experiences on my team.
2. I feel judged in STEM spaces for what I wear.
3. I get anxious that others do not believe in my technical ability.
4. I have faced negative discrimination because of my gender.
5. I have seen negative discrimination towards others who share my gender.

Fig. 10. 2024 Gender differences in Bias, Gender Identity and Authenticity in the STEM Environment



1. I feel authentic to myself in STEM spaces.
2. I feel judged for what I wear in STEM spaces.

3. I have experienced negative discrimination in STEM spaces because of my gender.
4. I have experienced negative discrimination in STEM spaces because of other factors.

Fig. 11. 2025 Gender differences in Bias, Gender Identity and Authenticity in the STEM Environment

The second dimension in both 2024 and 2025 is driven by feelings of perceived discrimination and judgement. Gender plays a significant role in these experiences, as can be seen in the results encapsulated by figures 10 and 11. In 2024, feelings of gender-based discrimination ( $p = 0.044$ ) and anxiety that others do not believe in their technical abilities ( $p = 0.028$ ) were statistically significant, with non-men reporting more experiences of gender discrimination and anxiety ( $M = 2.75$ ,  $SD = 2.125$ ,  $M = 4.5$ ,  $SD = 1.76$  respectively) compared to men ( $M = 1.83$ ,  $SD = 1.548$ ,  $M = 3.51$ ,  $SD = 1.938$  respectively).

In 2025, all survey questions in this dimension yielded statistically significant results ( $p < 0.001$ ). Non-men report feeling less authentic to themselves and more judged for what they wear ( $M = 5.59$ ,  $SD = 1.11$ ,  $M = 2.64$ ,  $SD = 1.35$  respectively) compared to men ( $M = 6.15$ ,  $SD = 0.99$ ,  $M = 1.95$ ,  $SD = 1.03$  respectively). Non-men also reported experiencing more discrimination both related to their gender and other factors ( $M = 4$ ,  $SD = 1.93$ ,  $M = 3.66$ ,  $SD = 1.61$  respectively) compared to men ( $M = 1.94$ ,  $SD = 1.45$ ,  $M = 2.75$ ,  $SD = 1.80$  respectively). Of significance is the mean difference of 2.06 when comparing the effects of gender on gender based discrimination. It should be noted that the questions in this dimension, especially those regarding discrimination, yielded high standard deviations, revealing a significant variation of experiences, even between those of the same gender.

The 2024 results indicate that non-men feel significantly more anxious about being judged for their technical ability than men do. This may lead to non-men opting for non-technical roles, where they might feel less judged and more confident in their abilities. These sentiments were once again reported by 2025 survey respondents, reinforcing the prevalence of this issue.

*“Women tend to apply for non-technical roles as a pathway into technical roles” - Female Non-MNR Engineering Team Member (2025)*

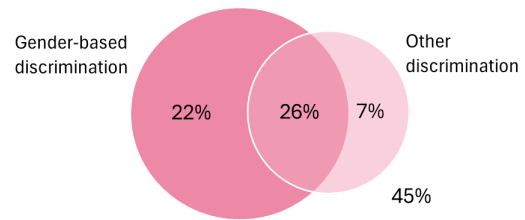


Fig. 12. 2025 Venn Diagram of non-men who reported experiencing discrimination in STEM spaces.

55% of all 2025 non-men respondents reported experiencing some form of discrimination in STEM spaces, with 48% of non-men respondents reporting gender-based discrimination, and 33% of non-men respondents reporting other forms of discrimination. Of the non-men who reported experiences of gender-based discrimination, 54% of them also reported experiencing non-gender-based discrimination. In contrast, of the non-men who did not report experiences of gender-based discrimination, only 15% reported other forms of discrimination. This indicates a strong intersectional overlap between gender based discrimination and discrimination from other sources in STEM environments. Further research should be conducted into this intersectionality in order to draw stronger conclusions. However, 55% of all 2025 non-men respondents reporting discrimination in STEM is significant in itself, reinforcing that discrimination in STEM is a considerable issue, and continued efforts are needed to help combat this. These results indicate that non-men are more likely to experience discrimination in STEM, and are less likely to feel comfortable in STEM spaces, with 2025 results reinforcing conclusions drawn in 2024.

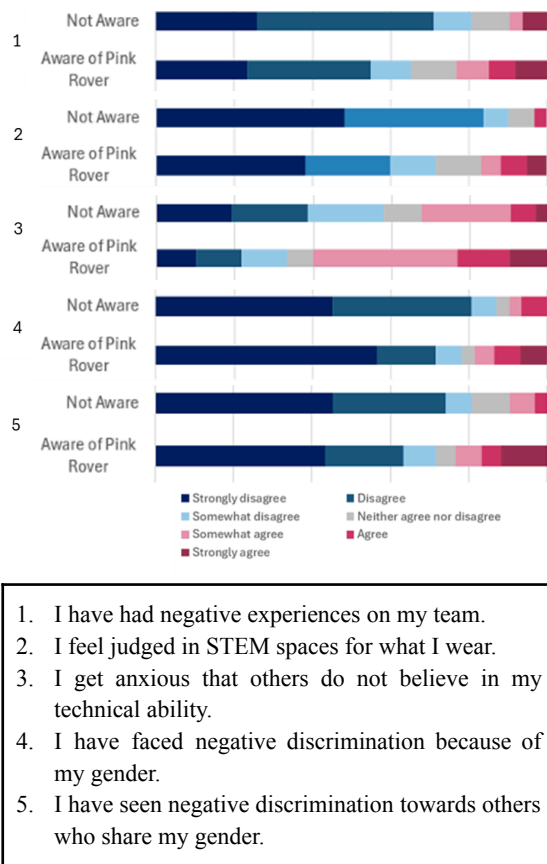


Fig. 13. 2024 Pink Rover Initiative Aware differences in Bias, Gender Identity and Authenticity in the STEM Environment

To explore the effect of exposure to the Pink Rover Initiative, an independent sample t-test was performed. The 2024 results reveal significant differences in the experiences of individuals involved in the Pink Rover Initiative compared to those not involved as seen in Figure 13. There were significant differences in feelings of anxiety and feeling judged, with both items yielding p-values of 0.014 indicating statistical significance. Participants who engaged with the initiative reported feeling more judged for their appearance and experienced heightened anxiety regarding their technical abilities ( $M = 4.28$ ,  $SD = 1.805$ ) compared to those not aware ( $M = 3.29$ ,  $SD = 1.755$ ).

2024 findings suggest that involvement in the initiative may correlate with increased awareness of social perceptions and pressures within STEM environments. 2024 participants who have faced gender discrimination or encountered negative experiences may be more inclined to engage with initiatives like Pink Rover, which are designed to support them.

In 2025, there were no significant differences relating to negative experiences in STEM when comparing participants who had and hadn't been exposed to the Pink Rover Initiative. This lack of findings in 2025 indicates that Pink Rover Initiative awareness is no longer a marker of negative experiences in STEM. Indeed, the 2025 survey data revealed that 77% of Monash students and 58% of non-Monash students were 'aware' of the Pink Rover Initiative, indicating that the initiative has reached a broader audience within the community. These findings suggest that initiative awareness has increased in those who do not face these experiences in STEM spaces; however, awareness does not ensure engagement. This is both a success that the initiative is able to reach a wider demographic in terms of awareness, and a limitation that it does not have a significant impact on everyone it reaches.

In 2025, comparison of other demographics revealed no statistically significant results related to negative experiences in STEM Environments; all significance was related to gender. This further highlights the large impact gender has in this dimension.

2024 interviews revealed that women felt their gender identity was strongly connected to their STEM pathway, whilst men did not share this sentiment. A common experience among women was feeling pressured to hide their authentic gender expression in order to be accepted within STEM spaces.

*"I didn't want to be seen as a girly girl." - Female Engineer. (2024)*

*"I suppress my personality and identity [in engineering environments] so I can be taken seriously." - Female Engineer (2024)*

These sentiments were supported by quantitative data in both 2024 and 2025, and highlight a concerning trend: the devaluation of femininity in STEM fields creates pressure for women to downplay their authentic selves in order to gain acceptance and credibility in male-dominated environments. This pressure not only stifles individuality but also perpetuates the notion that success in engineering requires conformity to masculine ideals, ultimately undermining the diversity and richness that different perspectives can bring to the field.

The Pink Rover Initiative prompted many women to reflect on their own unconscious biases, particularly in how they presented themselves and the time spent worrying about being accepted while suppressing aspects of their identity and self-expression, such as femininity.

*“Genuinely after the Pink Rover I realised I dialed up my masculinity to feel safe and comfortable and also preferred more masculine people... but dress is a big part of my gender identity and I realised I can express both femininity and masculinity.”- Female Engineer from MNR (2024)*

Many also noted the initiative prompted others to reflect on unconscious gender-based bias.

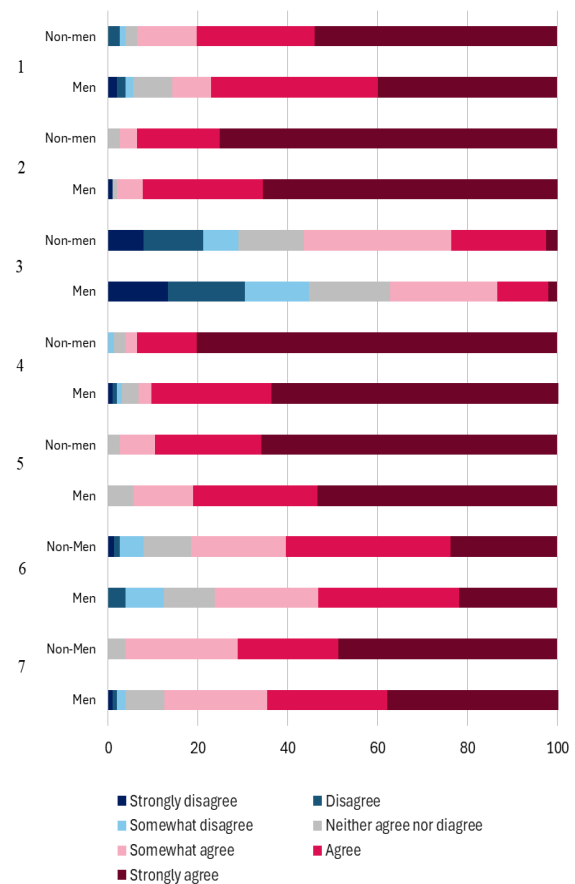
*“... using the colour pink in an area that is traditionally male dominated, actually does generate discussion.” - Female Engineer (2024)*

Prompting individuals to reflect on their own internal gender-based bias is critical to helping women feel comfortable expressing their authentic gender identity. Change is impossible without awareness of the biases one perpetuates, and thus discussions around the impact of an absence of femininity in STEM spaces must be held with all community members.

The initiative resonates strongly with women of various backgrounds, regardless of their feelings toward pink, as it made them reflect deeply on how they acted and the biases they placed on themselves and others. However, the initiative was limited in provoking self-reflection on unconscious bias in men and is a recommended area to be focused on in future.

*“Pink is not my thing but I understand what it represents.”- Female Engineer from MNR (2024)*

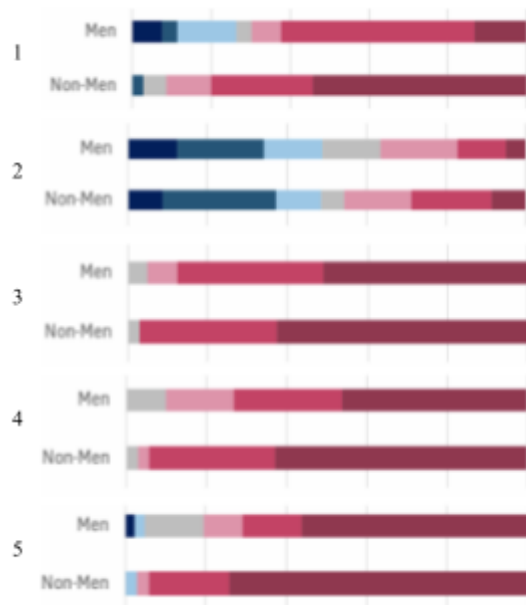
### 3.3.4 Allyship, Responsibility and Awareness



1. I believe that the underrepresentation of women in STEM is a problem.
2. I believe it is crucial that men are allies and support women in STEM.
3. I feel that expressing feminine identity is not accepted in STEM.
4. I believe that all genders should lead pursuits for gender equality in STEM.
5. I feel I have a role to play in creating inclusive environments for women.
6. I feel confident to start conversations around self-expression and women in STEM.
7. I feel motivated to champion diversity, equity and inclusion in STEM causes.

Fig. 14. 2025 Men vs Non-Men Differences in Allyship, Responsibility and Awareness

The statements above aim to measure Allyship, Awareness and Responsibility of issues relating to gender equity in STEM by analysing participants' understanding of expressing identity, fostering inclusive environments and sparking conversations about diversity.



1. I believe that the underrepresentation of women in STEM is a problem.
2. I feel that expressing feminine identity is not accepted in STEM.
3. I believe it is crucial that men are allies and support women in STEM.
4. I feel I have a role to play in creating inclusive environments for women.
5. I believe that all genders should lead pursuits for gender equality in STEM.

Fig. 15. 2024 Men vs Non-Men Differences in Allyship, Responsibility and Awareness

The 2024 results highlight significant gender differences in perceptions of gender equity in STEM as shown in Figure 15. Independent sample t-tests conducted in 2024, revealed that non-men were more likely than men to believe that the underrepresentation of women in STEM is a problem (Mean = 6.20 vs. 5.71, Cohen's  $d = 1.507$ ). Non-men also reported feeling that expressing a feminine identity is less accepted in STEM (Mean = 3.89 vs. 3.63, Cohen's  $d = 1.871$ ), indicating a strong divergence in experiences regarding gender identity. One notable result from the 2025 results showed an overall increase in the mean (Mean = 4.25 vs 3.64), although there was a drop in the effect size (Cohen's  $d = 0.369$ ). Regarding the 2024 results, non-men felt a greater responsibility for creating inclusive environments (Mean = 6.54) than men (Mean = 6.10), with a large effect size (Cohen's  $d = 0.888$ ), and expressed a stronger belief that all genders should lead gender equity initiatives (Mean = 6.63 vs. 5.98, Cohen's  $d = 1.203$ ). These results suggest that

non-men are more attuned to gender inequities and the need for inclusivity in STEM, though no significant gender differences were found in beliefs regarding the importance of men as allies. The two-way ANOVA does however reveal a significant gender-MNR interaction on D5 (who should lead gender equity efforts,  $p = 0.032$ ) and approaching significance for D4 (responsibility for inclusivity,  $p = 0.055$ ). This indicates that gender influences perceptions of leadership in gender equity efforts, with non-men placing greater emphasis on shared responsibility across genders, thus indicating the need to target men in these social campaigns and improve allyship recruitment.

The 2025 results, however, yielded limited gender differences in perceptions of gender equity. Independent samples t-tests showed that non-men were still more likely to report that expression of feminine identity is less accepted in STEM (Mean = 4.25 vs 3.64,  $p = 0.015$ ), but with a decreased effect size (Cohen's  $d = 0.369$ ). Other beliefs, such as recognising underrepresentation of women in STEM as a problem and the importance of men as allies had no significant gender difference. This change from 2024 to 2025 findings demonstrates an increase in men's perception of DEI efforts as a shared responsibility.

A two-way ANOVA test was conducted to further investigate the nature of such decrease in gender difference.

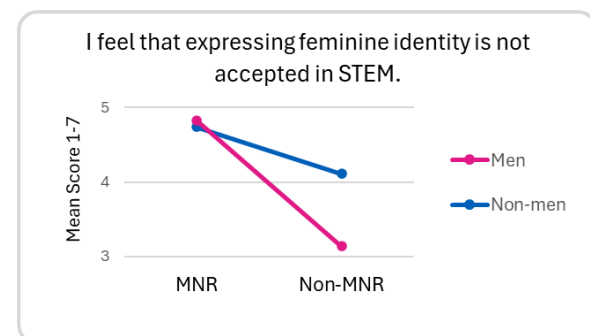


Fig. 16. 2025 Men vs Non-Men and MNR vs Non-MNR ANOVA mean differences for feeling that feminine identity is not accepted in STEM.

The two-way ANOVA of the acceptance of feminine identity in STEM revealed a significant MNR-gender interaction. The 2025 findings revealed that MNR men were the most likely to believe that expressing feminine identity is not accepted in STEM

(Mean = 4.83) while non-MNR men were more likely to believe that expressing feminine identity is more accepted in STEM (Mean = 3.14). The findings also revealed that MNR non-men were more likely to believe that expressing feminine identity is not accepted in STEM compared to non-MNR non-men (Mean = 4.74 vs 4.11). The significant gender effect and MNR membership effect suggests that the lack of overall gender difference found from independent t-tests is largely the product of perceptions shifting specifically for men within MNR. This can likely be attributed to the greater presence of the Pink Rover Initiative within MNR and fostering of a team environment that promotes conversations and awareness across genders.

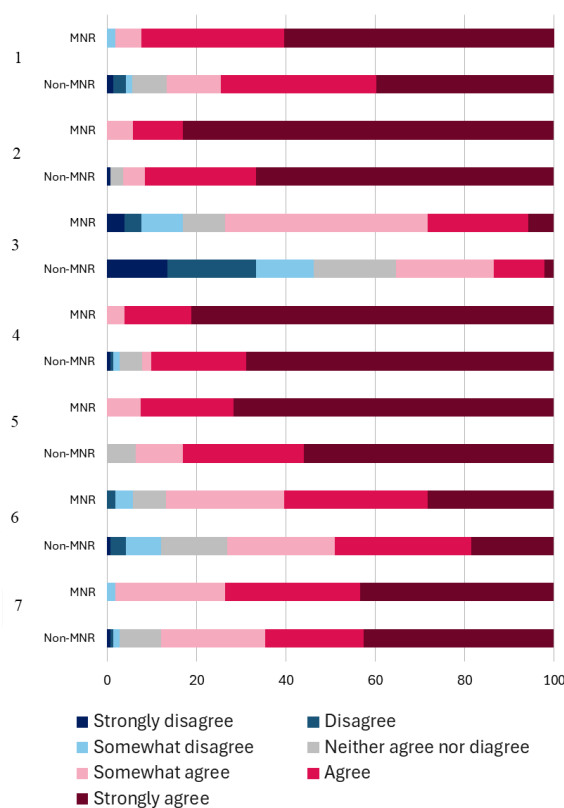


Fig. 17. 2025 MNR and Non-MNR Differences in Allyship, Responsibility and Awareness

To further analyse the impact of belonging to MNR, independent samples t-tests were performed and the results showed that individuals who were part of MNR were more likely to recognise underrepresentation of women in STEM as a problem (Mean = 6.49 vs 5.89,  $p < 0.001$ ), with a medium effect size (Cohen's  $d = 0.496$ ). The results also indicate that MNR participants are more likely to believe it is crucial that men are allies and support

women in STEM (Mean = 6.77 vs 6.52,  $p = 0.018$ ) with a relatively small effect size (Cohen's  $d = 0.317$ ). The 2025 findings also indicate that MNR participants feel that expressing feminine identity is not accepted in STEM (Mean 4.79 vs 3.58,  $p < 0.001$ ) with a relatively large effect size (Cohen's  $d = 0.754$ ). Moreover, the standard deviation for this statement was relatively high for MNR ( $SD = 1.364$ ) and higher for non-MNR ( $SD = 1.687$ ), which indicate that non-MNR participants have greater variability in their agreement with the statement. Results also show statistical significant MNR membership effect on belief that all genders leading pursuits for gender equality in STEM, with MNR individuals shown to be more likely to agree with this statement (Mean = 6.77 vs 6.46,  $p = 0.006$ ) with a relatively small effect size (Cohen's  $d = 0.332$ ). The findings illustrate that MNR individuals are more likely to believe that they have a role to play in creating inclusive environments for women (Mean = 6.64 vs 6.33,  $p = 0.007$ ) with a small effect size (Cohen's  $d = 0.376$ ). One notable remark is the standard deviation for this statement was relatively small with MNR data points being more clustered around the mean ( $SD = 0.623$  vs  $0.906$ ). These results indicate that belonging to MNR increases one's awareness around underrepresentation and gender identity in STEM, and their importance placed on shared responsibility and allyship in DEI efforts.

Furthermore, an independent samples t-test for MNR vs Non-MNR also revealed that MNR participants felt more confident to start conversations around self-expression and women in STEM (Mean = 5.68 vs 5.23,  $p = 0.039$ ) with a relatively small effect size (Cohen's  $d = 0.335$ ). This result indicates that participants involved in MNR tend to feel a stronger sense of responsibility in regards to bringing more awareness to gender equity issues in STEM.

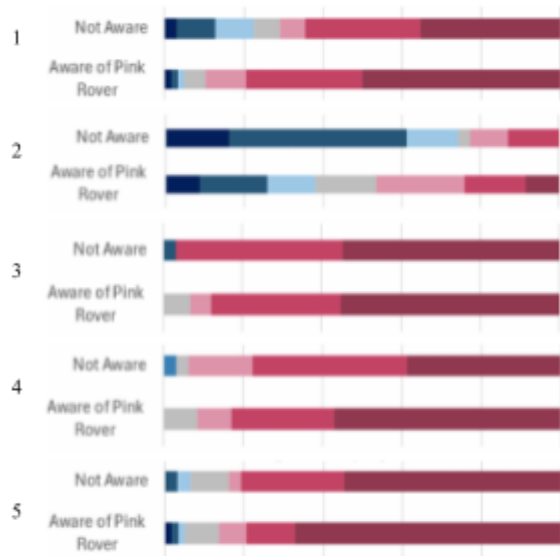


Fig. 18. 2024 Pink Rover Aware and unaware differences in Allyship, Responsibility and Awareness

The Pink Rover Initiative significantly impacted 2024 participants' views on gender representation and identity issues in STEM. Independent samples t-tests showed that individuals aware of the initiative were more likely to recognise the underrepresentation of women in STEM as a problem (Mean = 6.09 vs. 5.32,  $p = 0.014$ , Cohen's  $d = 1.529$ ) and express concerns about the acceptance of feminine identity (Mean = 4.07 vs. 2.84,  $p = 0.001$ , Cohen's  $d = 1.755$ ). These results indicate that the initiative successfully raised awareness around underrepresentation and gender identity. However, the initiative did not significantly influence beliefs about men's role as allies ( $p = 0.388$ ), responsibility for creating inclusive environments ( $p = 0.110$ ), or who should lead gender equity initiatives ( $p = 0.339$ ). The two-way ANOVA analysis corroborates these findings, showing that Pink Rover Initiative awareness significantly affected beliefs about underrepresentation ( $p = 0.043$ ) and identity ( $p = 0.001$ ), but had no significant impact on other gender equity dimensions (D3-D5).

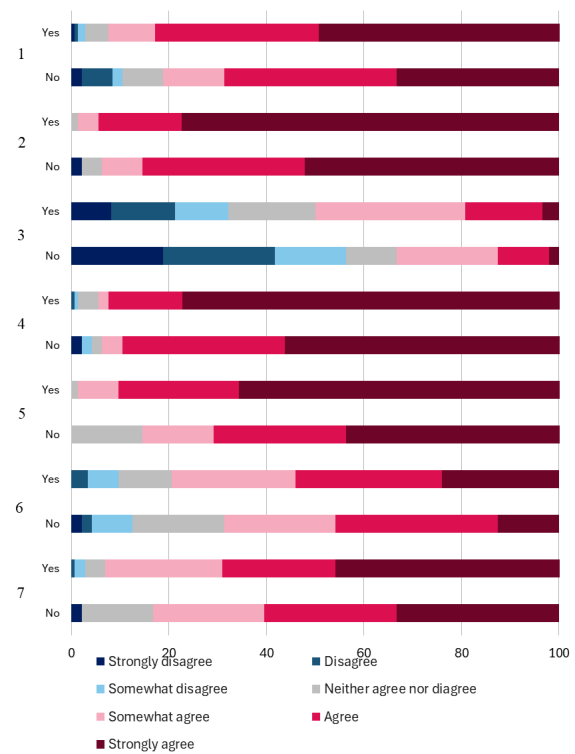


Fig. 19. 2025 Pink Rover Aware and unaware differences in Allyship, Responsibility and Awareness

The 2025 Pink Rover aware vs unaware aware results show statistically significant results. The participants who were aware of Pink Rover were more likely to believe that the underrepresentation (Mean = 6.20 vs 5.63,  $p = 0.02$ ) of women in STEM is a problem with a medium effect size (Cohen's  $d = 0.475$ ). The results also showed that Pink Rover aware participants were more inclined to agree that men should support women in STEM (mean 6.71 vs 6.25,  $p = 0.009$ ) with a medium effect size (Cohen's  $d = 0.593$ ). Interestingly, the standard deviation for Pink Rover aware participants was 0.612 as opposed to 1.12 for unaware. These results indicate that Pink Rover aware participants were more consistent in their agreement with this statement. Pink Rover aware participants were also more likely to believe that expressing feminine identity is not accepted in STEM (Mean = 4.11) compared to the unaware participants who tended to somewhat agree (Mean = 3.31,  $p = 0.004$ ) with a medium effect size (Cohen's  $d = 0.48$ ). There was quite a large standard deviation for both Pink Rover aware and unaware (SD = 1.619 vs 1.776) indicating a spread of values for the participants. The findings also revealed that Pink Rover aware individuals were slightly more likely to believe that all genders should lead pursuits for gender equality in STEM (Mean = 6.62 vs 6.31,  $p = 0.049$ ), however,

while the data was statistically significant, there was a relatively low effect size (Cohen's  $d = 0.33$ ). The data showed that Pink Rover aware participants were more clustered with their responses with a lower standard deviation ( $SD = 0.864$  vs  $1.151$ ). Another notable findings show that Pink Rover aware participants were more likely to feel that they have a role to play in creating inclusive environments for women (Mean =  $6.55$  vs  $6.00$ ,  $p = 0.002$ ), with large effect size (Cohen's  $d = 0.671$ ) and clustered responses ( $SD = 0.705$  vs  $1.092$ ). Thus, while Pink Rover aware and non-Pink Rover aware participants agreed that they have a role to play in creating inclusive environments for women, Pink Rover aware participants tended to agree with this statement more than unaware participants. These results indicate that participants who were aware of the Pink Rover Initiative were more likely to recognise gender inequity in STEM and be more aware of the issues surrounding lack of diversity.

The analysis of awareness of gender inequalities in STEM (2024 PC4) shows that awareness of the Pink Rover Initiative had a significant effect, highlighting its role in raising consciousness about gender disparities. However, gender itself did not significantly impact this awareness, and the interaction between gender and initiative awareness was also insignificant. In contrast, Allyship and Responsibility (2024 PC5) showed no significant effect from initiative awareness, although gender approached significance, suggesting that perceptions of allyship and responsibility may differ slightly between genders. As indicated, the initiative's impact was more pronounced in fostering awareness of gender inequities than in driving allyship and responsibility and is an area to target.

The 2024 quantitative data suggests that some men may not see advancing women in STEM as their responsibility. However, in the interviews, both men and women strongly agreed that everyone shares the responsibility to address these issues and had a deep understanding of gender inequality. The fact that all 2024 participants were familiar with the Pink Rover Initiative suggests it may play a role in fostering a sense of responsibility towards gender equality in STEM.

*"Without question, there's always more we can do—ensuring fair representation, elevating women, supporting women in engineering initiatives, and*

*fostering mentoring programs..."— Male Engineering Professor (2024)*

Compared to the 2024 findings, the 2025 findings show an overall increase in the mean, but a decrease in the effect size. In regards to Pink Rover aware vs unaware, 2025 participants were shown to be more likely to be aware of underrepresentation of women in STEM.

Similarly, the 2025 results showed an increase in awareness in regards to the acceptance of feminine identity in STEM space from Pink Rover aware vs unaware participants (Mean =  $4.11$  vs  $3.31$ ,  $p=0.004$ , Cohen's  $d = 0.48$ ). One notable difference is that the 2024 findings had a greater effect size (Mean =  $4.07$  vs  $2.84$ ,  $p = 0.001$ , Cohen's  $d = 1.755$ ).

The 2024 findings reveal notable gender differences in perceptions of equity in STEM, with non-men more likely to recognise the underrepresentation of women and the challenges of expressing feminine identities. The Pink Rover Initiative has effectively increased awareness of gender disparities, particularly regarding women's underrepresentation and identity acceptance. While the 2024 findings succeeded in highlighting these issues, it did not significantly alter beliefs about men's roles as allies or responsibilities in creating inclusive environments. The 2024 analysis indicates that gender influences perceptions of leadership in gender equity efforts, underscoring the importance of targeting men in future initiatives to enhance allyship and collective responsibility for fostering inclusivity in STEM.

The 2025 results reveal notable differences in perceptions of equity in STEM, with Pink Rover aware individuals being more likely to recognise the underrepresentation of women, the challenges of expressing feminine identities, and the potential roles that people can take to create inclusive environments. While there was little significance, in regards to the role that gender plays in influencing participants', the overall means increased which indicates that men and non-men are generally more aware of gender equity issues in STEM. The 2025 findings also yielded significant differences for MNR and non-MNR, where MNR participants were found to be more aware of equity issues in STEM and more willing to take steps to address these issues.

### 3.3.5 Response to Pink Rover and Gender Symbolism

The proposal of the Pink Rover Initiative sparked significant discussions within MNR, highlighting tensions around gender symbolism, representation, and the team's image. Despite initial concerns, the team's clarity of message and vision ultimately shifted perceptions about the initiative's potential impact.

2024 Survey results revealed varied initial reactions among team members. Of the 17 MNR participants who completed the survey, all voted in favor of the Pink Rover Initiative. However, 69.3% expressed skepticism when they first heard about it. This shift from skepticism to unanimous support suggests that the Pink Rover Initiative successfully addressed initial doubts, indicating a broader cultural change within the team. Additionally, all but 2 of the 48 MNR participants in the 2025 survey indicated that, if the team were to create a new rover they would be in support of the rover remaining pink, and no responses indicated opposing a pink rover, with the remaining two responses indicating that they were impartial to a change in future iterations of MNR's flagship rover. This finding suggests that the importance of Pink Rover to the team has been maintained since the launch of the initiative, with the core values represented by the initiative becoming integral to MNR team identity.

Once Pink Rover was launched, most 2024 interviewees reported that the initiative was well-received by the broader community and did not recall significant backlash.

*"My little cousin saw the pink rover on ABC Behind the News and that left a huge impact on her."*  
- 2024 survey participant

However, a few individuals expressed strong opposition to the initiative. For instance, at the SXSW event in Sydney, one MNR member shared that.

*"One woman voiced that she found the Pink Rover to be tokenistic, claiming it undermined women's progress."* - Male MNR Member (2024)

This reflects a concern that using pink to represent women in STEM may reinforce reductive or outdated gender stereotypes, which some feel trivialises the serious work of women in these fields.

In contrast, reactions at the Avalon Airshow revealed a different kind of bias. One person remarked,

*"What's up with the girl rover?" - Avalon Airshow Attendee (2024)*

This remark implies that the presence of the Pink Rover was perceived as inherently being gendered in nature. This comment could also further suggest an underlying discomfort in associating femininity, symbolised by the color pink, with technical fields like engineering, revealing a bias that sees STEM as inherently non-feminine. These examples highlight two different forms of bias: one criticises the initiative for being overly simplistic in its representation of women, while the other reflects discomfort with the integration of traditionally feminine symbols in male-dominated spaces like STEM. Both responses reveal the polarising nature of the initiative and the entrenched societal biases around gender representation.

These biases were less prevalent in 2025 survey results, although multiple respondents indicated having conversations around virtue signalling or tokenism as opposed to effective changemaking in relation to Pink Rover. One respondent wrote that conversations within their team were

*"Mostly about the PR that MNR gets due to the initiative."*

Other responses expressed a lack of clarity around the initiative's approach to making change,

*"none of us were sure as to what the initiative did apart from the panel event and the rover branding being pink"* - Female from non-Monash University Australian student team (2025)

This remark may be indicative of weaker promotion or clarity of the initiative for further removed teams (i.e. external to MNR, or external to Monash University). It also suggests that the referenced aspects of the initiative are insufficient action to address the issues faced by women in STEM, calling for more tangible impact.

Multiple other respondents reported conversations being centred around the design itself.

*“The conversations were around the colour rather than WISTEM” - Gender-fluid non-Monash University Australian Student Team Member (2025)*

*“it was very eye grabbing” - Male MNR Member (2025)*

*“the pink rover looks very flashy and fun, and overall brings a great vibe” - Man from non-Monash University Australian student team (2025)*

This trend shows that the core messaging and principals of the initiative are sometimes unclear or overlooked, but these responses also encompass several positive reactions to the design, demonstrating acceptance and even enthusiasm for “pretty”, traditionally feminine designs in STEM. This shows both a success in the initiative's ability to normalise what is often seen as atypical expression in STEM spaces, and a persisting limitation in being discounted by some as tokenistic or otherwise struggling to convey substantial meaning to select audiences. Improving clarity and visibility of this messaging to external parties may allow the initiative to combat this limitation and further convey meaningful ideas.

The quantitative 2024 survey questions provided valuable insights into participants' changing perspectives during the launch of Pink Rover. One participant noted,

*“The early discussions were uncertain about the direction and focus of the campaign... while the campaign may not tackle every gender-related issue in STEM, it serves as an excellent conversation starter that can lead to broader discussions about equality and diversity.” - MNR Engineer (2024)*

Engaging in conversations about women in STEM proved pivotal for many. The initiative's purpose is not to provide all the answers, but to spark reflection and dialogue. Participants appreciated the safe spaces created for diverse perspectives, allowing for complex discussions about the initiative's meaning. Ultimately, the Pink Rover Initiative succeeded in promoting conversations about women in STEM and encouraging thoughtful reflection.

Initial resistance to the idea stemmed from concerns regarding gender symbolism, particularly around the fear of reinforcing stereotypes. One 2024 participant expressed:

*“I initially felt that using pink to represent women in STEM was overly reductive... Realising it was more about femininity being welcome in STEM rather than being connected to it.” - Gender Fluid MNR Engineer (2024)*

This reflection highlights that thoughtful execution of the initiative helped alleviate concerns about reinforcing stereotypes and demonstrated genuine messaging around women in STEM and the representation of more nuanced messages such as the coexistence of femininity with technical fields such as STEM.

The Pink Rover itself exists as a powerful symbol of MNR's commitment to diversity and inclusion. The initiative challenges traditional perspectives, encouraging participants to reflect on biases. The symbolic choice of the colour pink emphasises that femininity and technical competence can coexist without compromise. As one participant noted,

*“This campaign breaks down our traditional constructs in a positive way.”- Male Engineering Professor (2024)*

However, this symbolism is not universally recognised in initial reactions to the initiative. An interview with a Monash University staff member revealed a student's initial negativity about the Pink Rover, stating,

*“A student felt the pink was reductive, the barbie-fication undercuts the involvement of serious women in STEM.” - Woman from University Engineering Faculty (2024)*

This feedback prompted the staff member to explore the topic further and eventually advocate for the initiative, recognising its value as a conversation starter. Most interviewees shared a similar journey from skepticism to support in the stages prior to the launch of Pink Rover, viewing the Pink Rover as a tool to navigate complex issues. While pink cannot represent all women, using it thoughtfully can open conversations and make traditionally marginalised identities feel more welcome in STEM.

This observation was a key factor in evolving the Pink Rover Initiative's messaging to more explicitly focus on broad and nuanced experiences and groups through the phrase ‘self-expression’. It was expected

that this broader focus would be reflected in the conversations sparked by the initiative, however 2025 survey results primarily captured an enduring connection between the initiative and gender when participants were asked to describe conversations sparked by Pink Rover. Of the responses commenting on which demographics were focused on, a majority exclusively mention women or girls. Several more mention a non-specific or broader group of genders (“non-men”, “all genders”, “gender diversity and inclusivity”, etc), suggesting that some discussion has included gender diverse experiences and potentially men’s experiences. A select few named traits or groups not inherently related to gender, specifically “LGBTQ+ People”, “Self expression”, “expression [that is] pink or stereotypically girly”.

These responses indicate that the initiative was able to spark conversations beyond gender, but the dominating trend being an exclusive focus on gender suggests that these conversations are still much less prevalent. There is also an absence of any discussion specific to disability, ethnicity, or religion as aspects of identity or self expression in STEM.

These findings show that Pink Rover has successfully sparked conversations about women in STEM, with the newer goal of broader conversations about self-expression starting to see success. The introduction of these nuanced conversations can be strengthened by clear and consistent messaging, as well as continual engagement with community members.

Despite the limited presentation of symbolism beyond gender in qualitative trends, quantitative data from 2025 showed that participants scored highly on the likert scale (Mean=6.17, SD= 0.781) in agreement to the statement “*The Pink Rover Initiative has encouraged conversations about broader forms of discrimination in STEM, beyond gender, and positively influenced my sense of inclusion and engagement*”. This result demonstrates that while the connection between gender and the initiative is the primary emphasis for many participants, broader conversations are being stimulated, with positive impacts on inclusion and engagement achieved.

Several 2025 participants also indicated a noticeable gender imbalance in who engaged in conversations sparked by Pink Rover, predominantly

reporting that women were more involved. A range of responses also commented on men’s engagement,

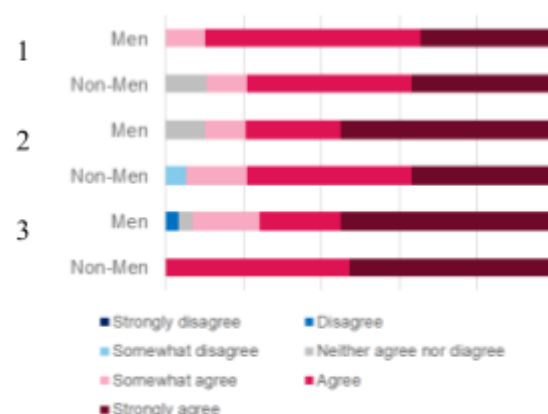
*“People who were most affected by [inequality and biases in STEM] contributed the most.” - Male MNR Member (2025)*

*“I think there were many male identifying members that were unsupportive and did not understand the significance of highlighting women in male-dominated spaces.” - Female on non-MNR Monash University team (2025)*

*“Generally women are more engaged with men’s initial beliefs challenged the most. Especially around the colour itself” - Male MNR Member (2025)*

*“Mainly the women were engaged the most, but several other male members also contributed to the conversation.” - Male on non-MNR Monash University team (2025)*

These responses indicate that fewer men than non-men actively engage in conversations, but several men do engage even if initially unsupportive. This demonstrates that Pink Rover successfully enabled open conversations with unsupportive parties, acting as a platform where individuals feel comfortable in voicing concerns or differing opinions. The recurring sentiment within MNR that men are more likely to initially express scepticism, while MNR showed no gender disparity in internal support for the initiative, exhibits the initiative's success in garnering support from and promoting open mindedness among men.



1. I support Pink Rover because I strongly relate to the key messaging and mission: to spark conversations about self-expression in STEM.
2. I feel that I can express my concerns about the Pink Rover Initiative.
3. The Pink Rover Initiative has encouraged conversations about broader forms of discrimination in STEM, beyond gender, and positively influenced my sense of inclusion and engagement.

Fig. 20. 2025 Gender differences in response to pink rover and gender symbolism within MNR

An independent T-test comparing MNR men to MNR non-men, showed there was not a statistically significant difference between the responses relating to support of Pink Rover and comfort with its core messaging. Both groups yielded means above 6.00 on the likert scale for each question on feeling able to express concerns about the initiative relating to messaging, and positive influence on engagement. This suggests that there is no persisting disparity between men and non-men on MNR regarding perspectives on inclusivity in STEM.

A thematic analysis was conducted on 2025 participant responses to the question “What attracted you to join Nova?” in which responses were compared according to themes of

- Interest / Passion (space, robotics, chemistry, “looked cool”, etc)
- Practical experience / applying skills
- Social aspect (“community”, “friends”, “social life”, etc)
- Culture, diversity/inclusivity

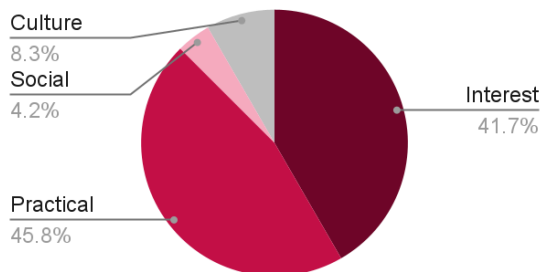


Fig. 21. Men's Response Themes, “What attracted you to join Nova?”, 2025 survey

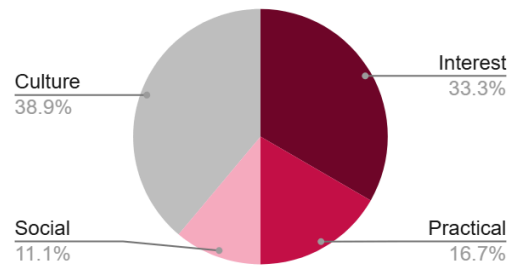


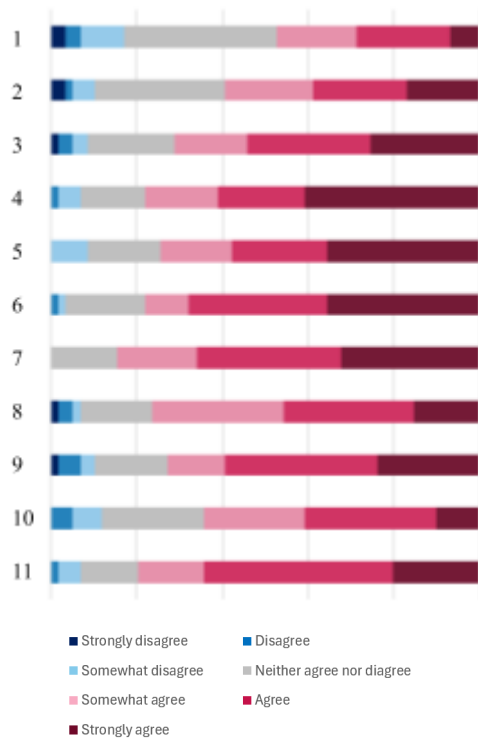
Fig. 22. Non-Men's Response Themes, “What attracted you to join Nova?”, 2025 survey

The aim was to identify differences in priorities for student team experience between men and non-men. Results show a significantly more prominent emphasis on team culture and inclusivity for non-men MNR members as opposed to men MNR members who tended to emphasise practical engineering experience. Furthermore, three women and one non-binary participant referred to Pink Rover specifically when discussing the team culture, while no men included any mention of the initiative. This supports both the notion that an inclusive environment is a highly important factor for non-men in STEM, and that Pink Rover primarily impacts non-men in terms of personal engagement in engineering.

Pink Rover has proven to serve as a catalyst for discussions about representation, identity, and progress in STEM. While initial reactions to gender symbolism were mixed, the thoughtful execution of the initiative prompted reflection and dialogue among team members. This highlights the need for sensitivity and awareness in navigating gender issues in STEM. Pink Rover has demonstrated profound impact, particularly through sparking conversations about women in STEM, as well as other gender identities. The initiative has also started to see success surrounding other aspects of identity and self expression, with this facet being an area of improvement for the future of the initiative.

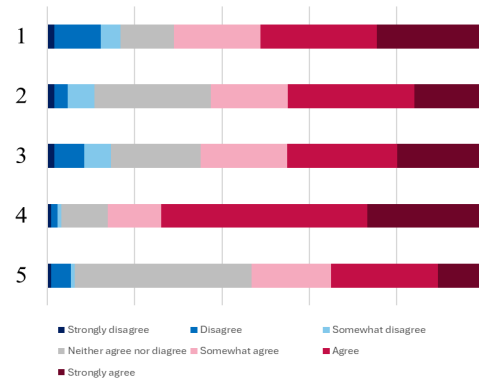
The primary impacts of Pink Rover are observed as predominantly distinct for men and non-men, in that the initiative has a greater impact on non-men's engagement in STEM, whereas the initiative has a greater impact on men's perceptions of STEM. Moving forward, the lessons learned from the Pink Rover Initiative can inform future initiatives aimed at promoting inclusivity and challenging stereotypes, ultimately fostering a more welcoming environment for all individuals in STEM.

### 3.3.6 Pink Rover Initiative Satisfaction



1. I observed other people reflected on unconscious bias.
2. I feel the concept of Pink Rover prompted me to self reflect on my unconscious bias.
3. I feel confident to start conversations around Pink Rover.
4. I feel comfortable advocating for the Pink Rover Campaign.
5. I feel comfortable defending the Pink Rover campaign.
6. I would strongly recommend Pink Rover events to girls and women I know.
7. I feel more girls and women would be more enticed to participate in space robotics after attending Pink Rover events.
8. I felt Pink Rover gave a platform to individuals to share their experiences in STEM openly.
9. I felt Pink Rover was successful in reinforcing belonging, acceptance and self-expression.
10. I felt Pink Rover was successful in highlighting stereotypes and bias in STEM.
11. I felt Pink Rover was successful in starting conversations about women in STEM.

Fig. 23. 2024 Pink Rover Satisfaction - Average Scores across all participants



1. Seeing the Pink Rover sparked conversations within my team.
2. I have observed that the Pink Rover Initiative prompted other people to reflect on unconscious bias.
3. I feel that the concept of the Pink Rover prompted me to reflect on my unconscious bias.
4. The Pink Rover Initiative promotes the importance of everyone having a sense of belonging in STEM.
5. I feel that the Pink Rover Initiative has helped create an inclusive and comfortable environment for me.

Fig. 24. 2025 Pink Rover Satisfaction - Average Scores across all participants

Surveys results from both 2024 and 2025 reveal several interesting trends that highlight its impact. Despite the absence of statistically significant differences in demographic comparisons between men and non-men, the shared sentiments across both 2024 and 2025 reflect a collective recognition of the initiative's importance in starting conversations about women and self-expression in STEM and challenging stereotypes. This consensus highlights the potential for the Pink Rover Initiative to unite various stakeholders in promoting diversity and inclusion. Results from 2025 indicate a particularly high mean score for the promotion of everyone having a sense of belonging in STEM; a shared attitude across all participants, including men, non-men, MNR members and other national and international participants. Furthermore, the overall high mean scores across years indicate a positive reception and acceptance of the initiative within the community over time as seen in Figures 23 and 24.

Table 5: 2025 Pink Rover Initiative Satisfaction Mean Scores, MNR and non-MNR, for questions listed in Figure 24.

|    | Group   | Mean | Std. Deviation | Cohen's D |
|----|---------|------|----------------|-----------|
| 1. | MNR     | 5.67 | 1.071          | 1.172     |
|    | Non-MNR | 4.36 | 1.139          |           |
| 2. | MNR     | 6.25 | 0.717          | 0.673     |
|    | Non-MNR | 5.53 | 1.263          |           |
| 3. | MNR     | 5.98 | 1.049          | 1.12      |
|    | Non-MNR | 4.46 | 1.517          |           |
| 4. | MNR     | 5.71 | 1.045          | 0.795     |
|    | Non-MNR | 4.66 | 1.458          |           |
| 5. | MNR     | 6.06 | 1.103          | 0.981     |
|    | Non-MNR | 4.59 | 1.704          |           |

All five statements included in the 2025 survey are of statistical significance ( $p < 0.001$ ) when a comparison is made between MNR and non-MNR individuals, and have Cohen's d values that indicate a medium or large effect. All means are higher for MNR than non-MNR, indicating that internal impact and initiative satisfaction are significant, with external impact and initiative satisfaction amongst other 2025 participants (other Monash University teams, other Australian and international teams) less so. However, non-MNR participants show more variability than MNR participants, with larger SD values for each of the five survey questions in this dimension. This indicates a larger spread of results beyond MNR, suggesting that opinions related to the initiative are more mixed within the broader community.

As the initiative was started by MNR members to support MNR members, with the highest exposure and involvement by MNR members, these results correlate with a higher internal impact. However, high mean scores from non-MNR respondents indicate that external initiative impact is still positive; this is both a success and a limitation of the Pink Rover Initiative, with external satisfaction and engagement an area of future improvement.

These insights could be instrumental in refining future initiatives, ensuring they resonate effectively with diverse audiences while continuing to challenge

existing biases in the industry. Overall, the data not only illustrates the initiative's current effectiveness but also opens avenues for further exploration into how different demographics perceive and interact with initiatives aimed at promoting equity in STEM.

Involving diverse perspectives in shaping the initiative is crucial to avoiding a single, dominant narrative. A more inclusive approach allows for the identification of potential blind spots and ensures the initiative resonates with a wide audience. By seeking diverse opinions and input from the community, the initiative promotes collaboration and fosters a sense of shared ownership, making people more likely to support and advocate for it. This collective involvement also strengthens the initiative's ability to challenge traditional gender norms effectively.

The 2025 short-answer survey questions were analysed using thematic analysis to identify areas of future improvement for the Pink Rover Initiative.

In response to 'What are some areas that the Pink Rover Initiative could improve on?', participants consistently commented on the importance of increasing initiative awareness and impact beyond MNR. One participant remarked that an area of improvement would be:

*Working with other student teams so that their leaders can promote the campaign and start more consideration of culture - and how our processes impact culture - within other teams. - Male Monash University Student Team Member (2025)*

Furthermore, many respondents noted the importance of events and broader community engagement.

*I think to have community events or even lectures to talk about how we can as leaders in our respective teams can improve inclusivity]. - Gender-Fluid Non-Monash University Australian Team Member (2025)*

Expanding on Pink Rover's presence in communities outside of MNR, could both increase the initiative's reach for individual impacts but also create structural impact by inspiring other organisations to champion diversity and inclusivity. 2025 survey results already show the beginnings of organisational impact, with reports of the kinds of conversations

sparked within other student teams. Several respondents from multiple non-MNR student teams indicated that Pink Rover had prompted discussion around *“Whether we create our own initiative”*, particularly amongst the teams’ leadership or upper management. One participant even reported a clear eventuation of these discussions,

*“The pink rover initiative inspired our team to start having discussions around gender diversity and inclusivity within our team, and how we could too champion these initiatives. Recently, this has culminated in the development of our mascot” - Non-MNR Monash Student Team Member (2025)*

These reactions to Pink Rover demonstrate the viability of significant impact in wider communities, further justifying a greater focus on external engagement and outward facing messaging in future.

Another essential finding was the frequency of references to the desire for increased promotion via social media; almost half of the respondents referred to this as an area of improvement. One participant remarked that an area of improvement would be:

*Expanding its reach on a wider platform such as social media - Female MNR Member (2025)*

It was not only the frequency of social media promotion that was referenced; in fact, the clarity of the initiative’s goals was also noted by multiple participants, aligning closely with findings from 2024 interviews relating to the communication of nuanced goals. In the 2024 interviews, it was revealed that while the message of ‘pink’ as a symbol of defiance against gender norms was clear, some felt the initiative’s deeper objectives - such as addressing unconscious bias and promoting inclusivity - were not communicated as strongly. This lack of clarity may have hindered its ability to connect with all target audiences effectively, underscoring the need for clear, consistent communication in addressing sensitive issues like gender bias in STEM. The perpetual nature of this area of improvement across 2024 and 2025 data recommends it as a key focus for future initiative refinement.

Interviews conducted in 2024 also reveal the success of specific events, such as the 2023 Pink Rover Women in STEM Night. Participants consistently praised the event for elevating

underrepresented voices and fostering community engagement. One male interviewee remarked,

*“There were comments from women on stage raising the additional bias they had to overcome. I’ve heard these biases exist, but hearing personal stories made me think more deeply.” - Male Engineer from MNR (2024)*

This response highlights how personal accounts of gender bias can deepen understanding, especially among those less familiar with the unique challenges women face in STEM. Events like these are critical for creating a more inclusive culture by providing spaces for open dialogue.

*“[The Pink Rover] encourages girls to join Nova Rover... but if I was looking for a student team I would gravitate towards that [Pink Rover]. Hopefully it encourages other student teams to consider their practices and more broadly other universities” - Female, Non-binary Engineer from MNR (2024)*

The initiative’s success is also attributed to the presence of women in leadership roles at the time of its launch. As one recruit noted,

*“I thought it was so cool that we had two girls running the team.” - Female Engineer from MNR*

However, the onus of driving DEI efforts should not rest solely on the shoulders of women. Men, too, must actively engage in leadership roles in promoting inclusivity to create a truly balanced and diverse team.

Indeed, the effectiveness of the Pink Rover Initiative is closely tied to leadership. Many 2024 interviewees expressed concerns that its impact was limited by the direction provided by its leaders. As a vehicle for social change, Pink Rover requires bold leadership with a clear vision. Without a well-defined roadmap, the initiative risks being perceived as superficial or stagnating in its efforts to address complex issues like equality and representation in STEM. Strong leadership and community input is essential for driving the initiative forward, ensuring that it remains meaningful and transformative.

The team’s technical excellence as one of the world’s top rover teams reinforces the message that femininity and engineering are not mutually

exclusive. The focus, however, is not just on winning competitions but on fostering an environment where all team members can grow into accomplished engineers, ensuring that opportunities are accessible to everyone.

Ultimately, in both 2024 and 2025, the Pink Rover Initiative sparked vital conversations about uncomfortable topics, fostering a culture of openness and reflection. It not only built a sense of community within MNR, but resonated with other students within the broader community. The initiative's success, while marked by some challenges, illustrates its potential to drive meaningful change in promoting gender equity in STEM and can only be improved by actively seeking and incorporating feedback from diverse perspectives.

### 3.4 Pink Rover Impact - Success and Limitations

The above analysis and conclusions are summarised in Table 6 and answers the primary research question; what are the key factors that contribute to the Pink Rover Initiative's successes and limitations? Recommendations for future iterations of the initiative are also discussed.

Table 6: Two way ANOVA test interactions observed in principal components

| Section | Dimension | Two way ANOVA Test Interactions |                               |
|---------|-----------|---------------------------------|-------------------------------|
|         |           | Gender-PRI Interaction effect   | Gender-MNR Interaction effect |
| 3.3.2   | 2024 PC1  | No                              | N/A                           |
|         | 2025 PC1  | No                              | N/A                           |
| 3.3.3   | 2024 PC2  | No                              | N/A                           |
|         | 2025 PC2  | No                              | N/A                           |
| 3.3.4   | 2024 PC4  | No                              | N/A                           |
|         | 2024 PC5  | No                              | N/A                           |
|         | 2025 PC3  | N/A                             | Yes                           |
|         | 2025 PC4  | N/A                             | N/A                           |
| 3.3.5   | 2025 PC5  | N/A                             | N/A                           |
| 3.3.6   | 2024 PC3  | N/A                             | N/A                           |

Table 7: Statistical significance of t-test demographics observed in principal components

| Section | Dimension | T-Test Estimated Means    |                             |                |
|---------|-----------|---------------------------|-----------------------------|----------------|
|         |           | Statistically Significant |                             |                |
|         |           | Differences in gender     | Differences in PRI exposure | MNR vs Non-MNR |
| 3.3.2   | 2024 PC1  | No                        | Yes                         | N/A            |
|         | 2025 PC1  | Yes                       | No                          | Yes            |
| 3.3.3   | 2024 PC2  | Yes                       | Yes                         | N/A            |
|         | 2025 PC2  | Yes                       | Yes                         | Yes            |
| 3.3.4   | 2024 PC4  | No                        | Yes                         | N/A            |
|         | 2024 PC5  | Yes                       | No                          | N/A            |
|         | 2025 PC3  | No                        | Yes                         | Yes            |
|         | 2025 PC4  | No                        | No                          | Yes            |
| 3.3.5   | 2025 PC5  | No                        | N/A                         | N/A            |
| 3.3.6   | 2024 PC3  | No                        | N/A                         | N/A            |

### Key Successes

- Initial skepticism within MNR during the launch turned into near unanimous support, reflecting cultural change and effectiveness in discussing gender issues in STEM, with pink symbolism eventually embraced despite concerns about tokenism.
- The Pink Rover Initiative prompted many women to reflect on their own unconscious biases, particularly in how they presented themselves and the time spent worrying about being accepted while suppressing aspects of their identity, such as femininity.
- While there is a disparity between men and non-men's perspectives on inclusivity in STEM, this gap has largely been closed within MNR and correlates with high levels of engagement with the Pink Rover Initiative.
- Pink Rover has consistently acted as a catalyst for discussions and personal reflection on an individual level.
- The internal impacts of Pink Rover have led to more unified understanding of and connection to messaging within MNR at an organisational level.

- Conversations have successfully been sparked in engineering teams beyond MNR, as well as within the broader community.
- The Pink Rover initiative has gone beyond its goal to start conversations outside the MNR team but also inspire other student teams to embrace DEI values and start their own initiatives.
- While the connection between the Initiative and gender symbolism is strong, the Pink Rover Initiative successfully encouraged conversations about broader forms of discrimination in STEM beyond gender, and positively influenced a sense of inclusion and engagement.
- Incorporating diverse perspectives enhances the initiative's connection with wider audiences. It takes diversity to achieve successful diversity initiatives.
- Consistent messaging is crucial for promoting an understanding of the initiative's nuanced goals.
- Non-men showed stronger awareness of gender inequities in STEM and emphasised shared responsibility for inclusivity.

### 3.5 Limitations of the Research and Future Work

Research can only ever represent a model of the world, and it can only be hoped that it is as accurate as it is designed to be. The main limitations of this research will be within the survey design, recruitment of participants and timing.

Surveys are widely used as a data collection method, but they do have some limitations. Sampling bias can occur if the sample is not representative or if there is self-selection bias. There is the risk that individuals who have animosity towards the Pink Rover Initiative chose not to participate, skewing the results of the data collected. Response bias is another issue; the surveys rely on participants to answer questions truthfully, however, participants may opt for answers they perceive as more socially acceptable; the anonymity of both the 2024 and 2025 surveys was expected to mitigate this risk. With the 2024 survey results, there is the risk that temporal bias exists, as the survey was run one year after the launch of Pink Rover, thus participants may have diluted opinions and struggle to recall details from the beginning of the launch.

The 2024 survey also aimed to capture the perspectives of external stakeholders who had engaged with the Pink Rover Initiative during various events. This group included attendees from the women in STEM night, outreach workshops, and public showcases, whose insights could have provided valuable external viewpoints on the initiative's impact. However, due to challenges in effectively marketing the survey to this audience, the recruitment of participants from external stakeholder groups was limited. As a result, the survey data primarily reflects the opinions of Monash student team members, which narrows the scope of feedback and leaves a gap in understanding how the initiative is perceived by a broader public audience. A recommendation following the conclusion of the 2024

### Key Limitations

- The initiative was not largely effective in provoking unconscious bias self reflection in men; this area is recommended for future focus.
- Non-men reported lower levels of inclusion and technical confidence compared to men, indicating a need for increased support in team settings.
- Engagement levels were limited in men on DEI in STEM topics.
- The initiative's reach outside of MNR is limited by a lack of visibility or clarity of messaging as presented externally.
- The initiative did not significantly influence beliefs about men's roles as allies, highlighting an area for improvement.

### Key Insights

- Non-men reported lower levels of inclusion and technical confidence compared to men, indicating a need for increased support in team settings.
- Many women in the study reported feeling pressure to conform to masculine ideals, leading to discomfort and self-doubt.
- The initiative resonated more with women identifying with feminine traits, indicating the potential for enhancing feelings of inclusion and belonging in STEM.

### Recommendations

- Leadership for inclusivity should involve men actively participating alongside women.
- Strong leadership and a clear vision are essential for sustained initiative success.

data collection was that future efforts should focus on improving outreach strategies to ensure a more diverse and inclusive participant pool; the 2025 survey aimed to address this by expanding the participant pool to include engineering student team members from other Australian as well as International engineering teams. The aim of this survey design change was to gain further feedback from external perspectives, as well as capture initiative impact on students beyond MNR and beyond those in close proximity to MNR at Monash University.

If further evaluation of the Pink Rover Initiative was to be undertaken, conducting pre and post surveys at events and workshops would be recommended to capture change in bias, perception or experience. It would also be recommended to have an evaluation plan developed in parallel with the development of interventions and initiatives. This work could also be expanded to evaluate other initiatives and compare limitations and successes.

#### 4.0 Conclusion

The Pink Rover Initiative has successfully sparked conversations around self-expression and experiences in STEM, notably engaging non-men who report higher levels of discrimination and self-doubt. These participants engaged more deeply with the Initiative, leading to a significant reflection of unconscious bias regarding gender expression in STEM. The consensus among interviewees revealed that femininity is often devalued in STEM, highlighting the deep-rooted cultural challenges feminine people face.

While the Initiative generally raised awareness, it struggled to do so with men outside of MNR, who generally felt gender expression issues were less severe than non-men or their male MNR counterparts. Efforts within the initiative also struggled to increase feelings of responsibility and allyship in men.

This research demonstrates that whilst the Pink Rover messaging within MNR was clear, non-MNR respondents experienced more confusion around the aims and objectives of the Initiative. More targeted and consistent external efforts are needed to clarify the messaging and better help individuals outside of MNR to resonate with and benefit from the Initiative.

The Pink Rover symbolised defiance of gender norms and promoted inclusivity, although its

acceptance varied. Despite initial skepticism following the launch, it became a successful conversation starter, illustrating that imperfect symbols can initiate discussions and challenge stereotypes when used thoughtfully.

Future iterations of the Pink Rover Initiative should focus on actively engaging men, addressing responsibility directly, and developing inclusive strategies that extend beyond awareness-raising, especially in men outside of MNR. By leveraging these insights, the Initiative can continue to challenge gender norms and foster a more inclusive STEM environment, particularly for non-men, who reported lower feelings of inclusion and technical confidence.

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#### Appendix A 2024 Survey Questions

Note that all questions allowed for responses on a 7-point Likert scale unless otherwise indicated.

##### Student team experiences

- I enjoy working with my student team.
- I feel supported by my teammates.
- I feel strong confidence in my technical ability.
- I feel comfortable asking my teammates for help.
- I feel included and part of the team all of the time.
- I feel heard and listened to on my team.
- I feel I am able to contribute to the team.
- I feel valued on my team.
- I feel comfortable to make mistakes on my team.
- I feel trusted by my team mates.
- I feel authentic to myself in STEM spaces.
- I have had negative experiences on my team.
- I feel judged in STEM spaces for what I wear.
- I get anxious that others do not believe in my technical abilities.
- I have faced negative discrimination because of my gender.
- I have seen negative discrimination towards others who share my gender.

#### Pink Rover Campaign experience

- I felt Pink Rover was successful ...
  - In starting conversations about women in STEM.
  - Highlighting stereotypes and bias in STEM.
  - Reinforcing belonging, acceptance and self expression.
- I felt Pink Rover gave a platform to individuals to share their experiences in STEM openly.
- I feel more girls and women would be more enticed to participate in space robotics after attending Pink Rover events.
- I would strongly recommend Pink ROver events to girls and women I know.
- I feel comfortable defending the Pink Rover Campaign.
- I feel comfortable advocating for the Pink Rover Campaign.
- I feel confident to start conversations around Pink Rover.
- I feel the concept of Pink Rover prompted me to self reflect on my unconscious bias.
- I observed other people reflected on unconscious bias.
- Tell us some areas the Pink Rover Campaign could improve on. (Qualitative short-answer)

#### MNR Pink Rover Campaign Experience

- Check what applies to you. (Yes/No/NA)
  - Did you contribute to or attend any initial Pink Rover Campaign development meetings?
  - Did you express your opinions on the Pink Rover Campaign in the initial stages through the anonymous google feedback form?
  - When the Pink Rover vote happened, what did you vote for?
- I felt comfortable with the Pink Rover Campaign and supported the cause.
- I felt comfortable to share my views on the Pink Rover in the initial stages of development at team meetings.
- I was skeptical of the Pink Rover idea when I first heard about it.
- I felt included in the Campaign to the extent I wanted to be.
- I felt the Campaign leaders listened to feedback from others on the team.

- I supported Pink Rover because I strongly related to the key messaging and mission: To start conversations about women in STEM.
- I supported the Pink Rover because I wanted to support the people who had shared their experiences.
- I wasn't motivated to support the Pink Rover Campaign.
- I felt pressured to support the Campaign.
- Explain why you were hesitant or not hesitant to support the Pink Rover Campaign? (Qualitative short-answer)
- What changed your perspective on Pink Rover if it did? (Qualitative short-answer)

#### Perceptions of STEM

- I believe that the underrepresentation of women in STEM is a problem.
- I feel that expressing feminine identity is not accepted in STEM.
- I believe it is crucial that men are allies and support women in STEM.
- I feel I have a role to play in creating inclusive environments for women.
- I believe that all genders should lead pursuits for gender equity in STEM.

#### Appendix B 2024 Interview Questions

##### MNR Interview Questions

- Introduction
  - Hello, Thanks for meeting me today and doing this interview. I'm really excited to hear your opinions. First, state your name and tell me a little bit about yourself.
  - You have been asked to take part in this Monash University study. You have been provided an explanatory statement about this study. By undertaking this interview you understand what this research project is about and would like to participate.
  - Do you understand that being in this study is our choice and that you can change your mind and choose to not be part of this study anytime?
- Q1: How would you describe your journey into STEM?
  - How did you end up joining the MNR team?

- Q2: Have there been any interactions between your gender identity and your journey into STEM?
- Q3: What were your contributions or involvement in the Pink Rover Campaign?
- Q4: What were your first thoughts about the Pink Rover Campaign when you first heard about it?
  - Were you hesitant to support initially and if yes why? What kind of responses did you witness to the Pink Rover campaign within the team before the vote?
- Q5: How did you find the environment on the team to voice your opinions on Pink Rover before the vote?
  - Did you feel your perspective was heard and included before the final vote?
- Q6: At what stage did you feel your perspective of Pink Rover changed if it did?
  - How did your perspective change if it did?
- Q7: Did you feel motivated to support Pink Rover from the beginning?
  - Were you motivated by others or your own experiences to support or not support?
- Q8: Did the Pink Rover prompt you to self-reflect and think about unconscious bias?
  - What unconscious bias was revealed if any?
- Q9: How did Pink Rover change the way you thought? Or the way you act?
- Q10: What was the impact of having strong representation of women in leadership (75% executive and 50% technical and operational leads were women)?
- Q11: How do you think the Pink Rover Campaign has shaped/influenced the MNR team's culture?
- Q12: What kind of conversations did you have stemming from Pink Rover?
  - Who were these conversations with?
- Q13: Tell me about the Pink Rover after launch?
- Q14: Have you had any recent experiences with Pink Rover now? And if so, tell me about those.

- Q15: Was there backlash with the Pink Rover Campaign?
  - How were stereotypes impacted and were they reinforced?
- Q16: What do you see in the future for Pink Rover, is there anything that could improve it?
- Q17: Do you feel comfortable defending and advocating Pink Rover?
  - Have you ever had to defend or advocate for Pink Rover?
- Q18: Do you feel you have a role to play in improving representation in STEM?

#### External Stakeholder Interview Questions

- Introduction
  - Hello, Thanks for meeting me today and doing this interview. I'm really excited to hear your opinions. First, state your name and tell me a little bit about yourself.
  - You have been asked to take part in this Monash University study. You have been provided an explanatory statement about this study. By undertaking this interview you understand what this research project is about and would like to participate.
  - Do you understand that being in this study is our choice and that you can change your mind and choose to not be part of this study anytime?
- Q1: What were your contributions or involvement in the Pink Rover Campaign?
- Q2: What were your first thoughts about the Pink Rover Campaign when you first heard about it?
  - Were you hesitant to support initially and if yes why?
- Q3: At what stage did you feel your perspective of Pink Rover changed if it did?
  - How did your perspective change if it did?
- Q4: Did you feel motivated to support Pink Rover from the beginning?
  - Were you motivated by others or your own experiences to support or not support?
- Q5: Did the Pink Rover prompt you to self-reflect and think about unconscious bias?

- What unconscious bias was revealed if any?
- Q6: How did Pink Rover change the way you thought? Or the way you act?
- Q7: What kind of conversations did you have stemming from Pink Rover?
  - Who were these conversations with?
- Q8: What do you think was most impactful about Pink Rover?
  - What are some specific elements that led to that impact?
- Q9: How did people respond to Pink Rover that you saw?
  - How did you think Pink Rover impacted the students / attendees?
- Q10: Was there backlash with the Pink Rover Campaign?
  - How were stereotypes impacted and were they reinforced?
- Q11: Do you feel you have a role to play in improving representation in STEM?

### Appendix C 2025 Survey Questions

Note that all questions allowed for responses on a 7-point Likert scale unless otherwise indicated.

#### Student team experience

- Regarding my student team experience, I feel...
  - Trusted by my teammates.
  - Valued on my team.
  - Supported by my teammates.
  - Comfortable asking my teammates for help.
  - Heart and listened to on my team.
  - Strong confidence in my technical ability.
  - Authentic to myself on my team.

#### Experiences of STEM

- I feel authentic to myself in STEM spaces.
- I feel judged for what I wear in STEM spaces.
- I believe that the underrepresentation of women in STEM is a problem.
- I feel that expressing feminine identity is not accepted in STEM.
- I believe it is crucial that men are allies and support women in STEM.
- I feel I have a role to play in creating inclusive environments for women.

- I believe that all genders should lead pursuits for gender equity in STEM.
- I see diverse self-expression in STEM environments.
- I feel motivated to champion diversity, equity and inclusion in STEM causes.
- I feel confident to start conversations around self-expression and women in STEM.
- I have experienced negative discrimination in STEM spaces because of my gender.
- I have experienced negative discrimination in STEM spaces because of other factors.

#### Pink Rover Initiative Experience

- I feel that the Pink Rover Initiative has helped create an inclusive and comfortable environment for me.
- The Pink Rover Initiative promotes the importance of everyone having a sense of belonging in STEM.
- I feel that the concept of the Pink Rover prompted me to reflect on my unconscious bias.
- I have observed that the Pink Rover Initiative prompted other people to reflect on unconscious bias.
- Seeing the Pink Rover sparked conversations within my team.
- What were these conversations centered around and who engaged the most? (Qualitative short-answer)
- What are some areas that the Pink Rover Initiative could improve on? (Qualitative short-answer)

#### MNR Pink Rover Initiative Experience

- What attracted you to join Nova? (Qualitative short-answer)
- I support Pink Rover because I strongly relate to the key messaging and mission: to spark conversations about self-expression in STEM.
- I feel that I can express my concerns about the Pink Rover Initiative.
- The Pink Rover Initiative has encouraged conversations about broader forms of discrimination in STEM, beyond gender, and positively influenced my sense of inclusion and engagement.
- If the team were to create a new rover, are you in support of the next rover being pink?

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